Paleomagnetism of Hawaiian Lava Flows

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Abstract. Paleomagnetic studies of lavas from the island of Hawaii have established the following: (1) All lavas now exposed on the island are less than 1 million years old (Pleistocene). (2) The present absence of nondipole field components in the central Pacific area is a relatively permanent feature, having lasted at least several hundred thousand years. (3) During the last million years, the main dipole field has wobbled, with periods up to 10^6 years and with amplitudes greater than 15°. (4) The average orientation of the main dipole field during the last 1 million years was displaced 5° from the present rotation axis.

INTRODUCTION

Oceanic volcanic islands play a prominent role in many geological and geophysical hypotheses, and, because Hawaii is one of the more thoroughly studied of these islands, the magnetic properties of Hawaiian rocks are of special interest. Moreover, because volcanism on Hawaii has continued throughout the period of direct observatory measurements of the geomagnetic field at Honolulu, we have been able to make a direct evaluation of the accuracy of the paleomagnetic method for recovering past geomagnetic field directions from Hawaiian lavas [*Doell and Cox,* 1963]. A qualitative impression of this accuracy may be gained from the results for these historic flows reproduced in Figure 1. The data are tightly grouped about the present geomagnetic field direction and indicate an accuracy sufficiently high to permit a rather detailed analysis of geomagnetic field directions to be made for this region.

The paleomagnetism of Hawaii is of further interest because this island is located in the central Pacific Ocean. Secular variation of the geomagnetic field in this region has been unusually low during the time spanned by observatory measurements [*Fisk,* 1931], and the nondipole component of the field is also largely absent at present [*Cox,* 1962]. Our earlier paleomagnetic study of historic Hawaiian lavas indicated that these conditions existed at least as far back as about 1750 [*Doell and Cox,* 1963]. The present results, summarized in Figure 1, suggest that secular variation was anomalously small during earlier time intervals (note especially the Puna and Hamakua volcanic series data). At other times the geomagnetic field in the central Pacific Ocean appears to have undergone rather marked changes in direction.

The ages of the lava flows exposed on Hawaii have generally been regarded as ranging from late Tertiary to Recent [*Stearns and MacDonald,* 1946; *MacDonald and Davis,* 1956]. Because several 180° reversals in the main dipole field are recognizable in rock strata of these ages from many parts of the world [*Cox et al.,* 1964], the thick sections of many lava flows exposed on Hawaii appeared to be ideal sequences for studying transitions in geomagnetic polarity. As may be noted in Figure 1, such a study of polarity reversal transitions in Hawaii has been thwarted by the complete absence of reversely magnetized rocks. On the other hand, this important result has indicated that Hawaii is considerably younger than was previously supposed [*Doell and Cox,* 1961], and recent radiometric studies [*McDougall,* 1964; *Evernden et al.,* 1964] confirm this age assignment.

All of these interpretations depend on the assumption that the paleomagnetic directions represent ancient geomagnetic field directions in the central Pacific. A major part of the present study concerns experiments made to support this assumption and to test further the accuracy of recovering past field directions from paleomagnetic data. The interpretations also rest on assumptions concerning the age relations of the various flows sampled; these are considered in the discussion of the paleomagnetic data.

DISTRIBUTION OF SAMPLES

Samples for this study were collected from all volcanoes comprising the island of Hawaii with

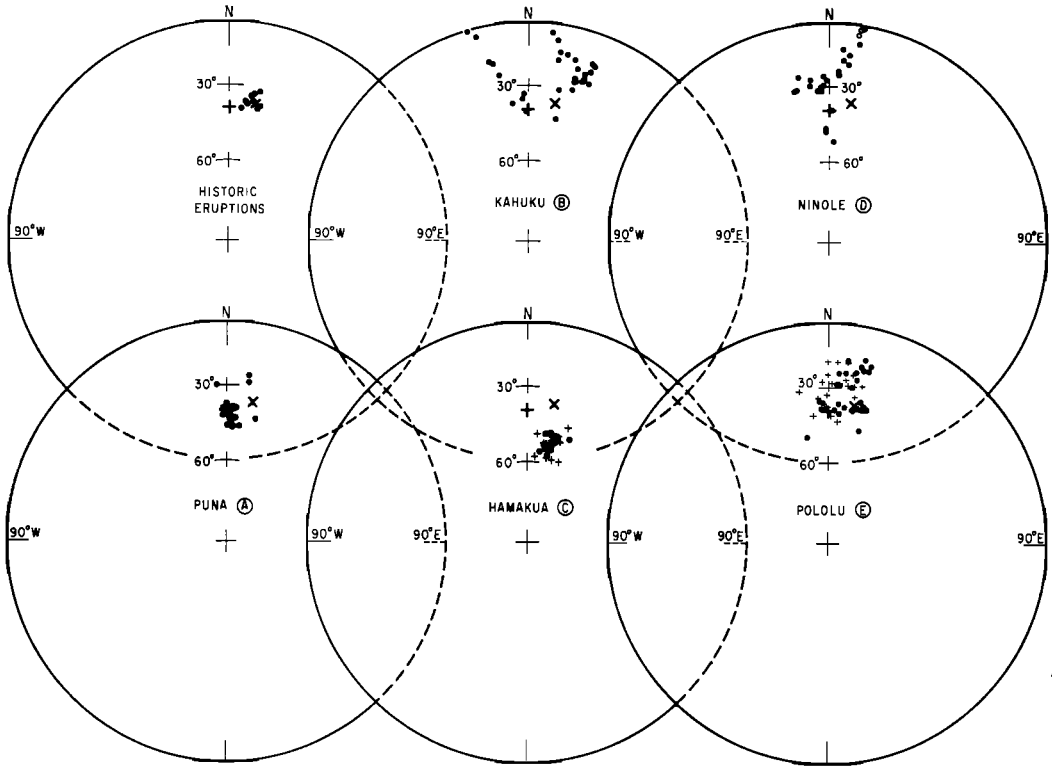


Fig. 1. Flow-average directions of NRM for Hawaiian lava flows. ●, north-seeking direction down; ○, north-seeking direction up, + axial dipole field direction in Hawaii, × inclined dipole field direction in Hawaii (+ 'single sample' directions, lower hemisphere (see Table 6)). Lambert equal-area projection.

the exception of Hualalai volcano; the sampling localities (Figure 2) were as much as 125 km apart. The age and stratigraphic summary in Figure 2 is taken from *Stearns and MacDonald* [1946]. The general stratigraphic relationships shown are based on the widespread beds of Pahala ash, considered as everywhere contemporaneous, and the fact that the two oldest sections, the Ninole and Pololu volcanic series, are separated from all younger series by a major erosional unconformity. Stearns and MacDonald regarded this erosional unconformity as separating Pliocene from Pleistocene rocks and cited a late Pleistocene age for the Pahala ash. Radiocarbon dates [*Rubin and Berthold*, 1961] on carboneceous ash from the top and bottom of the Pahala ash suggest that it was deposited between 10,000 and 17,000 years ago. Whether the lavas of the Kahuku and Hamakua volcanic series were extruded at the same time, as suggested in Figure 2, is not certain; nor is it at all clear that the sampled sections of the Ninole or Pololu volcanic series represent different periods of time. Details concerning the individual series and flows which were sampled are given in Table 6 and in the following remarks.

Pololu volcanic series. Probably the oldest flows sampled on Hawaii are those of the Pololu volcanic series, which are exposed on the southeast wall of Waipio Canyon. The complete section is described in detail by *Stearns and MacDonald* [1946, pp. 197–198], and the 138 meters sampled for this study are from the lowest portion. The flows nearer the top of the canyon appeared to be highly weathered and were not sampled.

Ninole volcanic series. Twenty-five flows, representing 56(+) meters of section were sampled from the northeast and southwest sides of Kaiholena Ridge. The two portions sampled are

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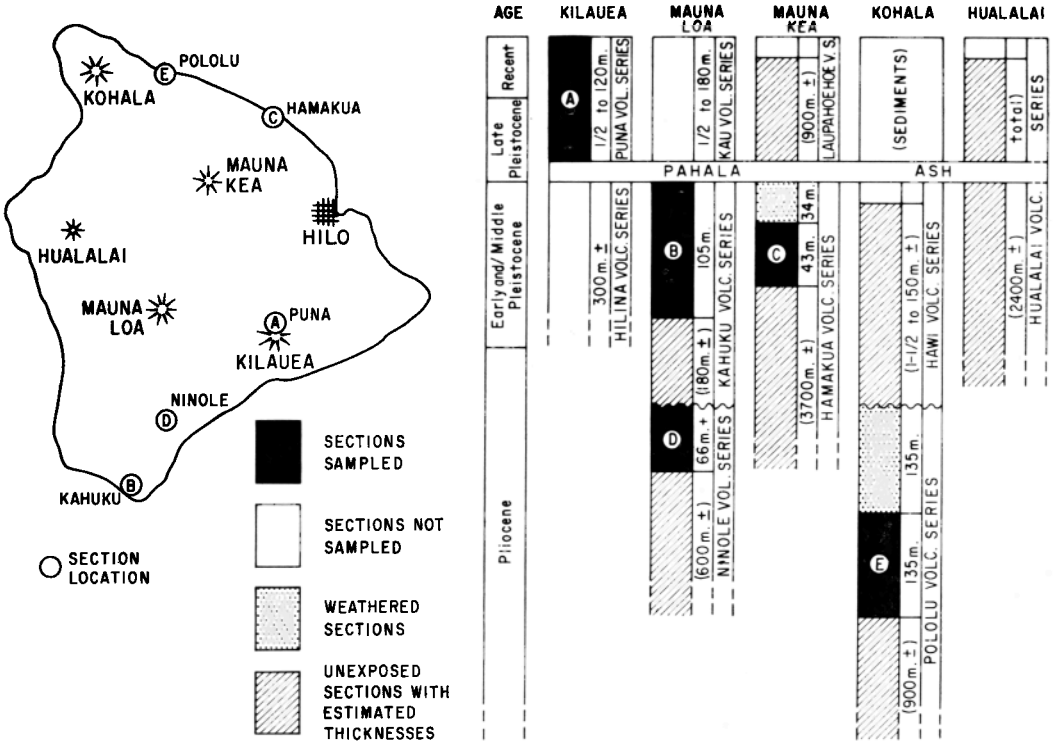


Fig. 2. Sampling site locations and generalized stratigraphic relations.

in fault relationship, and the number of flows originally interposed between those flows here numbered as units 66 and 67 is unknown.

Hamakua volcanic series. Twenty-six flows from the Hamakua volcanic series were sampled along the south side of Laupahoehoe Gulch; they are essentially the 'lower member of the Hamakua Volcanic Series' in the section measured at this locality by *Stearns and MacDonald* [1946, pp. 195–196]. The upper member is weathered and was not sampled.

Kahuku volcanic series. Lavas of the Kahuku volcanic series were sampled at the type locality, Kahuku Pali, near the southern point of Hawaii. Samples from 29 different flows making up 102 meters of section were collected. The uppermost flow sampled lies directly under the Pahala ash, and most of the flows exposed in the section were sampled.

Puna volcanic series. Eighteen flows, representing 131 meters of stratigraphic section of lavas of the Puna volcanic series, were sampled along the northern wall of Kilauea caldera (Uwekahuna Bluff). The section sampled is essentially that measured by *Stearns and MacDonald* [1946, pp. 193–194], and the two lowest flows sampled are separated from those above by an erosional unconformity.

A thick section of Kau volcanic series (post-Pahala ash) exposed in Mauna Loa caldera (Mokuaweoweo) was not sampled, nor was a thick sequence of Hilina volcanic series (pre-Pahala ash) exposed on Hilina Pali south of Kilauea volcano. The thin exposed portions of the Hawi, Hualalai, and Laupahoehoe volcanic series that mantle the Kohala, Hualalai, and Mauna Kea volcanoes were not sampled either.

All samples for these studies were taken in the form of cores 2.49 cm in diameter and from 10 to 20 cm in length, drilled in situ with a portable water-cooled diamond coring drill. A nonmagnetic orienting device was used to mark the cores before they were detached from the outcrop in such a way as to preserve the geographic orientation of each 2.28-cm-long specimen later cut from these cores for magnetic

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measurements. The accuracy of the orienting method is estimated to be ±2° (s.d.) [*Doell and Cox,* 1963].

Paleomagnetic Accuracy

The problem of paleomagnetic accuracy we take as the problem of determining how closely the average direction of remanent magnetization of a set of lava flows corresponds to the regional geomagnetic field direction at the time each flow cooled. In our earlier study of historic Hawaiian lava flows we showed that an accuracy of 3° to 5° was possible if five to eight samples were collected from each flow. However, the older Hawaiian lavas are several hundred thousand years older than the historic flows, and during this time some of the original magnetization may have been lost or new components of magnetization may have been added. Some of the older lavas were initially buried before being exposed by erosion, so that the possibility of remagnetization through reheating must be considered. It is also possible that some of the rather divergent directions observed (for example, the very shallow inclinations in the Kahuku and Ninole volcanic series, Figure 1) were influenced by magnetic anisotropy, so that the original magnetizations may not have been acquired parallel to the ambient magnetic fields at the time the flows cooled. The experiments described in this section were designed to evaluate the bearing of these factors on paleomagnetic accuracy.

It is generally not possible to distinguish original dip of lava flows from subsequent tectonic tilting; thus the older sections could have been tilted after acquiring their NRM. However, there is no geologic evidence for tectonic tilting of more than a few degrees on Hawaii.

Stability of NRM in alternating fields. Representative specimens from 33 different lava flows were partially demagnetized in alternating fields to determine whether the NRM (the natural remanent magnetization initially measured in the laboratory) was stable and to determine whether two or more components with varying degrees of stability were present in the NRM. (A description and evaluation of the demagnetization equipment used for these studies are given by *Doell and Cox* [1963; 1965*b*]; the induction magnetometers used for measuring remanent magnetization and an analysis of their operations are given by *Doell and Cox* [1963; 1965*a*; 1965*c*]).

The specimens were demagnetized in alternating fields with peak intensities, $\tilde{H}$, of 25, 50, 100, 200, 400, and 800 oe, and the percentage of the NRM remaining in each specimen after demagnetization at each field intensity $\tilde{H}$ is given in Table 1. Four representative demagnetization curves are shown in Figure 3, where $J_r(\tilde{H})/J_r(\tilde{H} = 0)$ is the percentage of NRM remaining after demagnetization in field $\tilde{H}$. The specimen from Pololu unit 17 is magnetically unusually 'hard' for a basalt, that from Ninole unit 55 is unusually 'soft,' and those for the other two are rather typical. Curves varying as widely as these were also found within the group of historic flows on Hawaii [*Doell and Cox*, 1963]. Among the present data there is no obvious relationship between magnetic stability in alternating fields and the age of the flows. Since the NRM of the historic flows was shown to be thermoremanent magnetization (TRM) acquired when they cooled, the similarity of the present demagnetization curves to those from the historic flows suggests that the NRM of the older flows is also mainly primary TRM.

The possibility remains that the NRM consists in part of an unstable component of magnetization small enough to go undetected in the demagnetization curves but large enough to have altered the original direction of magnetization. Such an unstable component might be added by the fields due to lightning strikes near the sampling sites [*Cox,* 1961; *Graham,* 1961], by the drilling or cutting operations, or by the addition of viscous magnetization resulting from long exposure to the weak field of the earth [*Rimbert,* 1956]. If such an unstable component is present, then on demagnetization in successively higher fields it will be selectively destroyed and the direction of magnetization of the specimen will move progressively toward the direction of the more stable TRM component.

In the demagnetization experiments on the 33 specimens, no systematic changes in the direction of magnetization were observed (Figure 4). Although occasional changes in direction of magnetization at fields of 400 and 800 oe were noted, when demagnetization experiments were repeated at these higher field values after changing the initial orientation of the specimen in the demagnetization apparatus, the direction of mag-

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TABLE 1. Percentage of NRM Remaining in Selected Flows after Partial Alternating Field Demagnetization at Peak Field Strength $\tilde{H}$

Unit No.	$\tilde{H}$, oe 25	50	100	200	400	800
	(*a*) Puna Volcanic Series					
145	101	100	97	89	54	24
143	100	100	96	83	46	19
138	97	91	82	62	36	16
136	99	98	92	77	49	22
132	95	89	74	52	30	18
	(*b*) Kahuku Volcanic Series					
129	97	91	73	46	23	9
122	95	86	79	57	31	14
121	95	92	82	60	32	14
119	96	94	91	78	58	28
114	94	91	77	55	34	21
107		99	90	62	27	14
103	101	98	95	85	49	22
	(*c*) Hamakua Volcanic Series					
99	99	97	90	68	43	24
94	98	93	87	66	45	25
91	100	98	93	73	41	23
85	91	71	55	39	16	3
	(*d*) Ninole Volcanic Series					
71	87	79	64	40	20	8
70	98	93	80	53	28	13
69	96	88	68	35	14	6
66		86	68	35	15	5
65	98	93	78	50	23	8
63	86	80	59	39	21	8
60	95	93	83	55	28	10
56	92	82	60	36	14	5
55	92	82	52	27	12	3
52	101	93	78	34	13	4
	(*e*) Pololu Volcanic Series					
49	96	86	78	53	20	6
42	100	99	97	57	44	15
26	100	94	75	43	21	7
17	100	99	98	91	63	29
14	99	99	89	48	16	6
11	100	94	79	56	31	17
2	99	88	68	52	30	11

netization changed on successive runs. This indicates that the observed changes are due to a component added by the demagnetization process [*Doell and Cox,* 1965*b*], rather than to the removal of a less stable component of magnetization. The absence of viscous and isothermal remanent magnetization is demonstrated by the striking coherence of directions of magnetization in fields up to several hundred oersteds.

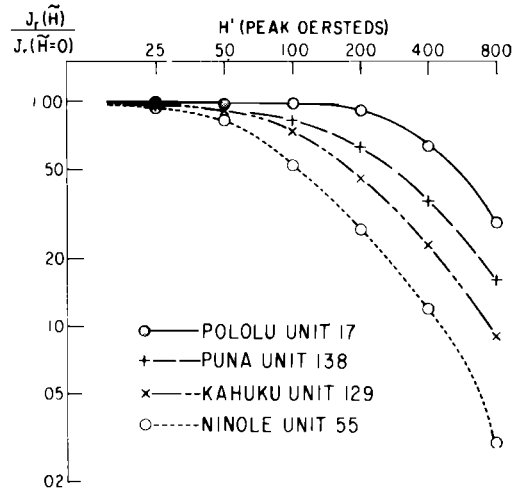


Fig. 3. Demagnetization curves for typical Hawaiian lavas. Ordinate is the amount of remanent magnetization, referred to the NRM, after treatment in alternating magnetic fields of peak field strength $\tilde{H}$.

Chemical change and remagnetization. Changes in remanent magnetization are known to accompany chemical changes that occur in ferromagnetic minerals after a lava flow cools; moreover, such chemical remanent magnetization may not be distinguished from TRM by ac demagnetization [*Haigh,* 1958; *Kobayashi,* 1959]. When collecting the samples, fresh cores were drilled from the centers of larger joint blocks so that material that had obviously been weathered was avoided. Thin sections were also prepared from every flow and an examination was made for alteration. No evidence of alteration after cooling was noted. However, this evidence is inconclusive because of the difficulty of distinguishing mineralogic changes occurring during cooling from those occurring afterwards.

The generally tight coherence of NRM directions between sample cores collected from different parts of individual flows (Table 6) also supplies indirect evidence that chemical alteration is absent or rare. Unless chemical alteration occurred simultaneously, and to the same degree, in all parts of a flow, which is unlikely, it would produce divergent directions of magnetization corresponding to different directions of the earth's field when the chemical changes occurred, provided that such geomagnetic field changes took place. This conclusion is strengthened by the observation that within each flow

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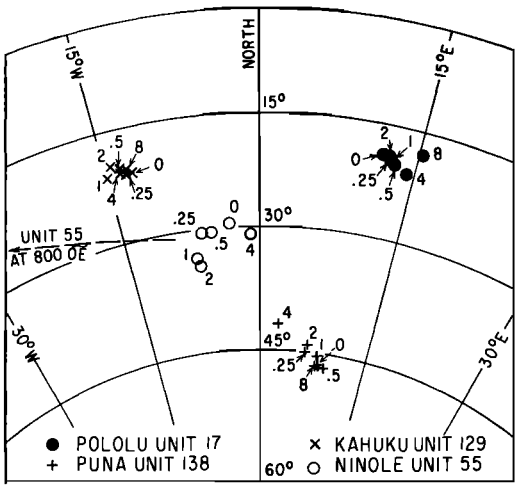


Fig. 4. Changes in direction of remanent magnetization during demagnetization for the four specimens shown in Figure 3. Values are peak alternating field strengths (oersteds $\times 10^{-2}$). Lambert equal-area projection.

there is considerable variation in opaque mineral grain sizes and habits and in the titanium content of the ferromagnetic minerals (Table 2); these factors greatly influence the vulnerability of the minerals to chemical change.

Curie temperatures and effects of reheating. A lava flow that is reheated to near its Curie temperature loses part of its original magnetization and on cooling may become remagnetized in a new direction. Where sequences of superimposed flows are sampled, as in the present study, there is always the possibility that any flow may have been reheated by the overlying flow and subsequently remagnetized. Elementary calculations indicate that unless an overlying flow is extremely thick, only a small part of the underlying flow will be heated sufficiently to be remagnetized, especially if the flows are separated by a layer of cinders, rubbly flow margins, or scoria, as is common.

In the present investigation, a considerable variation in flow thicknesses was noted, thicker flows often overlying thinner ones. If remagnetization by reheating has occurred, there should be a tendency for two such flows to have the same direction of magnetization. As may be seen in Table 6, this is not observed.

In sampling most flows it was possible to collect cores at different levels within the flow, although sampling immediately adjacent to the top was generally avoided. No tendency was observed for samples higher in the flows to have directions more nearly parallel to the overlying flows, again indicating that there has been little remagnetization due to heating by superjacent flows.

Thermal gradients are commonly very high in regions of active volcanism, especially near volcanic centers or vents, so that an entire section of flows buried to even moderate depth could be reheated and remagnetized; such remagnetization would be disastrous when an attempt is made to recover paleomagnetic information about secular variation. This possibility is especially great in flows containing titanomagnetites, which may have Curie temperatures of 250°C or less. The entire pile would cool much more slowly than an individual flow, and, since TRM is acquired over a temperature interval extending a few hundred degrees below the Curie temperature, the NRM of such lavas would not show the effects of secular variation due to an integrating effect.

Because a wide range in Curie temperatures is commonly encountered in basalts, the reheating of a pile of flows might act in another way to produce a false picture of past changes in the field. Consider, for example, the paleomagnetic data from the Pololu volcanic series (Figures 1 and 12), which display the following unusual characteristics. The oldest lavas are magnetized in a direction close to that of a geomagnetic axial dipole. The field then appears to have swung to a nondipole direction with inclination in the range 15° to 30° and declination in the range 0° to 15° east. Farther up in the section, there appear to be several more oscillations between these two directions. This may correspond to a real, if unusual, property of the geomagnetic field. However, the following model would also explain these data in terms of reheating. All of the flows may have cooled originally in a short time interval when the earth's field had the nondipole direction. Subsequently the pile was buried and reheated to, say, 300°C at a time when the field had changed to an axial dipole direction. Those flows with Curie temperatures less than 300°C were then remagnetized, which accounts for the apparent oscillation between the dipole and nondipole directions.

Data about Curie temperatures are clearly needed to test these possibilities. For this pur-

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pose, saturation magnetization as a function of temperature was measured on a continuously recording balance of the Faraday type for 63 rock specimens. Measurements were made in a vacuum of $<10^{-4}$ torr to inhibit oxidation. Temperature readings were calibrated to ±4°C using standard materials, and the magnet field strength and field gradients were calibrated to ±1% using coils suspended from the balance [*Doell and Cox,* 1965*d*]. These values may be taken as estimated accuracies for the Curie temperature and the magnetization data in Table 2.

Typical heating and cooling curves obtained with this balance are shown in Figure 5. Most of the heating and cooling curves are not strictly identical, which indicates changes in composition during heating. Where these changes are small, they appear to fall into one of two groups, (*a*) those where the cooling curve remains below the heating curve and (*b*) those where the cooling curve crosses the heating curve. For purposes of tabulating the data, the curves were further divided into the following categories. *Type 1*: Simple curve with single knee; ferromagnetic minerals are inferred to be magnetite or titanomagnetite of constant composition [*Akimoto,* 1955; *Nagata,* 1961]. *Type 2*: No well-defined knee; titanomagnetite grains with varying composition are inferred. *Type 3*:

TABLE 2. Results of Magnetic Balance Measurements

Unit No.	J_i, emu/gr	T_c, °C	Type
	(*a*) Puna Volcanic Series		
144	0.65	525	1*a*
143		550	1*a*
142(*a*)	0.46	525, 175	3
142(*b*)	0.53	550	1*a*
142(*c*)	1.07	550	2*a*
142(*d*)	1.04	550	1*a*
142(*e*)	1.74	525	1*a*
141	1.00	550	1*a*
140	1.51	550	1*a*
139	1.38	475	1*a*
132	0.60	500	1*a*
131	0.61	525	1*a*
	(*b*) Kahuku Volcanic Series		
127	0.79	475	1*b*
126	0.34	525	2*a*
125	0.34	500	1*a*
124	0.66	525	1*a*
123	1.24	525	1*a*
121	0.41	525	1*a*
120	0.19	550	2*b*
114	1.24	525	1*b*
113	0.22	550	2*a*
112	0.98	550	1*b*
104	1.27	500	1*a*
103	0.23	575	2*b*
	(*c*) Hamakua Volcanic Series		
99	1.28	550	1*a*
95	1.48	550	1*a*
94	1.26	550	1*a*
91	1.56	550	1*a*
90	0.83	525, 125	3
89	2.60	550	1*a*
88	2.18	525	2*a*
86	2.69	550	2*a*
85	1.52	525	1*b*
84	1.98	525, 175	3
82	0.23	575	2*b*
79	1.23	525	1*a*
	(*d*) Ninole Volcanic Series		
75	1.22	525	2*b*
74	1.52	(550)	4
73	1.12	500	2*b*
72	0.99	(550)	4
71	1.83	350	1*b*
70	1.41	525	1*a*
69	0.92	550	1*a*
60	0.62	550	1*b*
59	1.09	550	1*a*
58	0.95	500	1*a*
57	0.61	550, 250	3
53	1.47	500	1*a*
52	0.96	500	2*b*
	(*e*) Pololu Volcanic Series		
50	0.79	550	1*a*
42	1.62	550	1*a*
39	1.08	550, 225	3
33	0.90	525	2*b*
32	0.94	(500)	4
17	0.72	525	1*b*
16	0.65	(525)	4
15	1.09	550	1*a*
14	0.82	(500)	4
13		(500)	4
9	0.65	500, 225	3
8	0.56	550	2*b*
7	0.68	550	1*b*
6	0 73	550	1*b*

J_i, 'Saturation magnetization,' measured at room temperature in a field of 10,000 oe.

T_c, Curie temperature(s).

Type, Type of thermomagnetic curve as illustrated in Figure 5.

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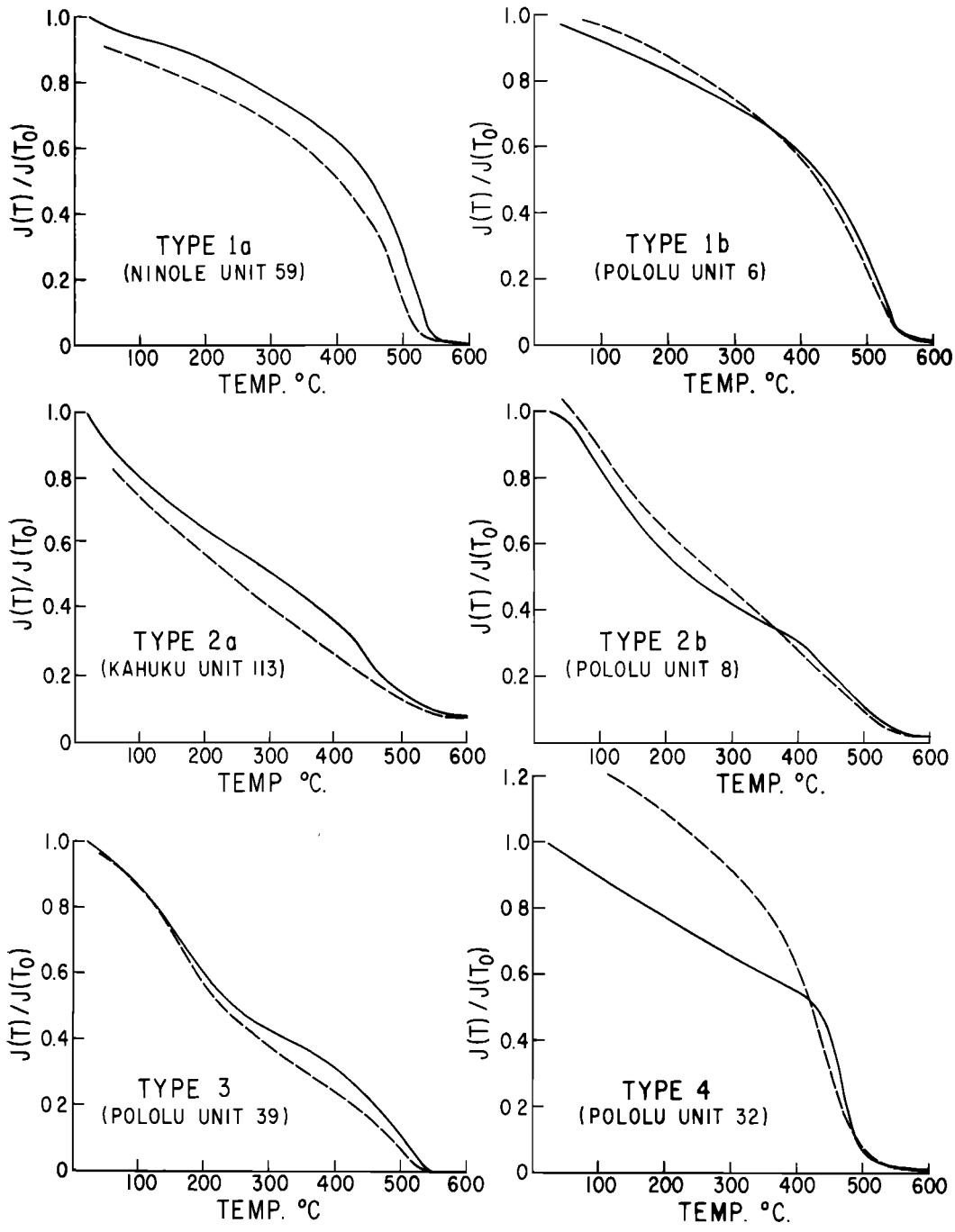


Fig. 5. Examples of the six types of saturation magnetization versus temperature behavior shown by the Hawaiian lavas. Ordinate is magnetization at temperature T (referred to magnetization at room temperature) in a magnetic field of 4000 oe; —— heating cycle, ------ cooling cycle.

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Two knees in curve; two distinct ferromagnetic mineral species are inferred (two Curie temperatures are given in Table 2). *Type 4*: Chemical alteration during heating is so great that the original Curie temperature cannot be determined; Curie temperatures in Table 2 (parentheses) refer to altered minerals after heating. In type 3 curves the (*a*) and (*b*) response is also apparent, but the distinction is not always clear owing to the presence of two different species undergoing different types of response.

To determine intraflow variation, specimens from all five cores of Puna volcanic series unit 142 were measured (Table 2). The variation within this flow, both in types of curve and value of saturation magnetization, was about as large as that found within most of the series. It is thus clear that any geologic correlation of flows based on these parameters would be unsound for basalts like those on Hawaii.

Since the Hawaiian lavas generally contain ferromagnetic minerals with Curie temperatures greater than 500°C, it is unlikely that more than a very few have been remagnetized after being reheated by superjacent flows. It is equally unlikely that entire sections have been remagnetized during burial.

We now consider the dual grouping of NRM directions in the Pololu volcanic series lavas. In Figure 6 the thermomagnetic properties and directions of NRM are plotted together. It is clear that these data do not support the previously suggested model for explaining the dual groupings in terms of reheating. Type 4 curves, which are characteristic of chemically unstable titanomagnetites with high titanium content and low original Curie temperatures, are found in specimens corresponding to both groups of directions, as are type 1 curves, which are characteristic of low titanium content and high Curie temperature titanomagnetites. These thermomagnetic data show that remagnetization by reheating has been unimportant in the development of the NRM of the Hawaiian lava flows.

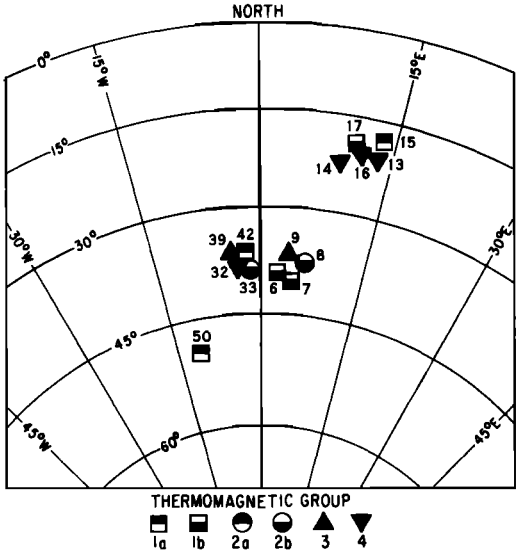


Fig. 6. Relations between NRM directions and type of thermomagnetic behavior for lavas of the Pololu volcanic series. Lambert equal-area projection.

Magnetic anisotropy. *Uyeda et al.* [1963] have shown that the direction of thermoremanent magnetization acquired by anisotropic rocks lies in a direction between that of the ambient field during cooling and the nearest direction of maximum susceptibility anisotropy. Directions of magnetization which are horizontal or even negatively inclined are found in some of the Hawaiian lavas, notably those from the Kahuku and Ninole volcanic series; the question arises whether these represent rather unusual excursions of the geomagnetic field or whether they are due to magnetic anisotropy in a horizontal plane caused by, for example, flow structures.

Anisotropy in the Hawaiian lavas was investigated by means of a microscopic examination of thin sections and polished sections of specimens from every flow, and by magnetic experiments. From the thermomagnetic measurements it is known that magnetite and titanomagnetite are the dominant ferromagnetic minerals in these rocks, and from the work of *Uyeda et al.* [1963] it is clear that the anisotropy of these minerals is an anisotropy of shape rather than of preferred crystallographic orientation. Attention was thus focused on visually identifying preferred orientations of elongate or planar shapes. Except for rather local alignments in parts of some sections, no systematic shape anisotropy was noted.

A susceptibility bridge [*Girdler*, 1961] or torque meter [*King and Rees*, 1962] is commonly used to measure anisotropy in paleomagnetic studies. However, these instruments only measure the anisotropy in domains with low or intermediate coercivities, whereas in lavas the

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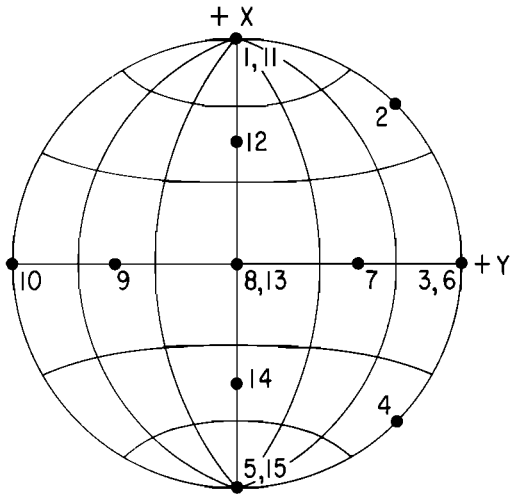


Fig. 7. Directions of inducing magnetic field within specimen for anisotropy experiments (+z axis down). Lambert equal-area projection.

larger part of the NRM is confined to domains with high coercivity. We have preferred, therefore, to use a method which measures anisotropy in domains with high coercive force [*Cox and Doell,* 1965].

In this method, the specimen is given an isothermal remanent magnetization (IRM) by applying a 7-kilogauss field along a precisely known direction. After this IRM has been measured, the specimen is demagnetized in an alternating field of peak strength $\tilde{H}$ and the remaining magnetization measured. This sequence is then repeated, applying the 7-kilogauss field along each of the 15 directions shown in Figure 7. For each remanent magnetization measurement we calculate a deviation vector $\mathbf{D} = \mathbf{J}_r - \bar{J}\mathbf{i}_H$, where $\mathbf{J}_r$ is the measured remanent magnetization produced in the field H, $\bar{J}$ is the mean amplitude of the set of 15 $\mathbf{J}_r$ vectors, and $\mathbf{i}_H$ is the unit vector in the direction of the field H. A set of 15 **D** vectors is obtained for each value of the demagnetizing field $\tilde{H}$. In our experiments we obtained a set of $\mathbf{J}_r$ vectors before demagnetization ($\tilde{H} = 0$), which shows the integrated effect of all domains, and a set after demagnetization at 800 oe, which involves domains with coercive forces greater than 800 oe. If anisotropy is present in either set of data, it will appear as a preferred alignment of the **D** vectors along easy directions of magnetization.

These anisotropy experiments were made on specimens from the Ninole volcanic series, units 71 and 70 (specimen numbers 1D156-1 and 1D161-1), both of which have unusually shallow inclinations. As a control, identical experiments were made on specimens from the Puna volcanic series, units 145 and 136 (specimen numbers 1D018-1 and 1D080-1); these units have typical inclinations close to the dipole field. No preferred alignments of **D** vectors were noted in any specimen, either before or after demagnetization at 800 oe, which indicates the absence of a detectable magnetic anisotropy.

The question then arises as to the minimum anisotropy that could be detected by these experiments. To evaluate this, the average value of the magnitude of the 15 deviation vectors **D** was calculated (Table 3). Because individual vectors are randomly oriented, the data in Table 3 may be regarded as estimates of the noise background, and an anisotropy would be detected only if it produced **D** values greater than these. From the values in Table 3 it appears that anisotropy greater than 1% could be detected for the entire coercive force spectrum, and greater than $2\frac{1}{2}$% for the part of the spectrum above 800 oe.

Anisotropy of this magnitude would produce angular deviations between the direction of TRM and that of the applied field of only 2° to 5°; thus the effects of anisotropy in the Hawaiian basalts is small, certainly much too small

TABLE 3. Results of Anisotropy Experiments

Specimen	$\tilde{H}$	$\bar{J}(\tilde{H})$	Average Amplitude of Deviation Vector (± s.d.)
1D018-1	0	1.78×10^{-1}	1.98 ± 1.01
	800	1.59×10^{-2}	5.70 ± 2.08
1D080-1	0	1.57×10^{-1}	1.37 ± 0.55
	800	3.34×10^{-2}	3.85 ± 1.41
1D156-1	0	6.19×10^{-1}	1.27 ± 0.71
	800	8.55×10^{-3}	3.32 ± 1.44
1D161-1	0	5.35×10^{-1}	1.84 ± 0.84
	800	2.08×10^{-2}	3.68 ± 2.25

$\tilde{H}$ is peak value of alternating field in oersteds.
$\bar{J}(\tilde{H})$ is average of intensity of IRM acquired in 15 directions in 7000-oe constant fields before and after demagnetization at 800-oe peak alternating field (in emu/cc).
Average amplitude of deviation vector given as per cent of $\bar{J}(\tilde{H})$.

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to account for the wider deviations of remanent magnetization directions.

Summary. The studies discussed above lead us to the following conclusions about the paleomagnetic reliability of these data: (1) The NRM measured in the Hawaiian lavas is predominately TRM acquired when the lavas cooled. (2) This NRM was acquired essentially parallel to the directions of the geomagnetic field at the time the lavas cooled. (3) The NRM of these lavas has not changed significantly since it was acquired. All these experiments indicate that the paleomagnetic accuracy of the results from the older Hawaiian lavas should be comparable with that obtained from the historic flows.

Paleomagnetic Results

In this section we will consider the problem of how best to recover and analyze information about the geomagnetic field obtained from the NRM measurements. There are several options in approaching this problem: multiple specimens from each core may be measured; demagnetization in alternating fields may be used to remove paleomagnetic noise due to soft components of magnetization imposed on the NRM; and any of several statistical schemes may be used in analyzing the measurements.

The time required for lava flows of the thickness of those studied on Hawaii to cool through the temperature interval during which they become magnetized ranges from a maximum of about 10 years to less than 1 year. Changes in the geomagnetic field direction during times this short are too small to be detected by paleomagnetic means. An important consequence is that each lava flow has sampled the geomagnetic field essentially at one point in time. Our attention will therefore first focus on the problem of recovering as much information as possible from each flow; analysis of data from groups of flows will be considered later.

Magnetic cleaning in alternating fields. Partial demagnetization in alternating fields may be used to obtain information of two types in paleomagnetic research. The technique may first be used as an analytical tool to determine whether two or more components of magnetization with different degrees of magnetic hardness are present in the NRM. Experiments for this purpose on specimens from 33 different flows were discussed above. The second use of partial demagnetization is in magnetically cleaning an entire set of samples from a given flow. The purpose here is to remove soft components of magnetization that may increase the angular scatter in directions of NRM, thus permitting the average direction of magnetization of the flow to be determined more precisely. The demagnetizing field strength to be used is determined by several considerations. If demagnetization experiments of the first type indicate that some minimum field is required to remove a systematic soft component, then that field should be used for the magnetic cleaning. Our experiments on the 33 specimens did not uncover a systematic soft component, and so there is no need for cleaning the Hawaiian lavas in such a minimum field.

Where no minimum field is required to remove a systematic soft component, the entire set of samples from one flow may all be demagnetized at progressively higher field strengths to determine the field strength at which angular scatter is minimized [*Irving et al.*, 1961]. We have done this, using alternating field strengths of 12.5, 25, 50, 100, 200, 400, and 800 oe, for one or two flows from each of the series sampled. The results are summarized in Table 4, where the angular standard deviation factor is a statistical estimate of the angular standard deviation of individual specimen directions of magnetization from the flow mean [*Cox and Doell*, 1964]. It should be noted that minimizing the angular standard deviation factor does not ensure that a systematic soft component has been removed. This must be established independently. Our previous experiments on 33 specimens indicated the absence of such a component; as a check on this we have calculated the average direction of magnetization for all flows after each set of demagnetization experiments (Table 4). These average directions do not change significantly at higher fields, which again indicates the absence of a systematic soft component.

Of the seven sets of experiments, only for Kahuku unit 114 is there a well-defined minimum in the values of ξ (angular standard deviation). For the other flows, values of ξ fluctuate at successively higher demagnetizing fields, all flows undergoing an increase in ξ after demagnetizing at 800 oe. The reason for both the random fluctuations of ξ and the increase at 800 oe

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TABLE 4. Angular Standard Deviation Factors and Average Direction of Remanent Magnetization for Selected Flows after Partial Alternating Field Demagnetization at Peak Field Strengths $\tilde{H}$

	$\tilde{H}$, oe							
	NRM	12.5	25	50	100	200	400	800
	(*a*) Puna Volcanic Series, Unit 132 (N = 9)							
ξ	0.141	0.135	0.139	0.147	0.144	0.154	0.148	0.174
D	7.5°	7.8	7.6	7.8	7.9	8.2	6.8	8.1
I	28.4°	28.4	27.5	28.0	27.8	27.9	27.9	29.2
	(*b*) Kahuku Volcanic Series, Unit 122 (N = 8)							
ξ	0.055	0.057	0.054	0.056	0.057	0.070	0.049	0.089
D	11.0°	10.3	10.5	10.3	10.8	10.7	10.5	11.0
I	18.0°	16.9	17.0	16.1	16.6	16.7	16.0	16.3
	(*c*) Kahuku Volcanic Series, Unit 114 (N = 7)							
ξ	0.095	0.076	0.072	0.063	0.079	0.084	0.083	0.149
D	16.7°	17.2	17.4	17.7	17.7	17.5	16.7	16.2
I	24.6°	24.0	24.2	24.2	24.2	23.8	24.3	24.8
	(*d*) Hamakua Volcanic Series, Unit 94 (N = 7)							
ξ	0.058	0.053	0.065	0.060	0.055	0.057	0.065	0.088
D	16.2°	17.1	16.9	17.9	17.4	18.0	18.6	18.8
I	55.9°	56.2	56.3	57.8	57.5	57.6	57.4	57.1
	(*e*) Ninole Volcanic Series, Unit 55 (N = 11)							
ξ	0.063	0.062	0.060	0.060	0.063	0.070	0.115	0.283
D	354.4°	353.8	353.4	352.8	352.0	352.3	349.7	348.0
I	29.4°	30.4	30.7	30.4	30.9	31.2	31.6	35.2
	(*f*) Pololu Volcanic Series, Unit 26 (N = 6)							
ξ	0.148	0.149	0.141	0.145	0.141	0.144	0.169	0.378
D	5.5°	4.9	4.3	4.6	3.7	4.8	0.8	22.7
I	18.6°	18.8	18.7	17.0	17.5	17.1	16.4	21.4
	(*g*) Pololu Volcanic Series, Unit 14 (N = 6)							
ξ	0.043	0.050	0.048	0.044	0.046	0.060	0.082	0.408
D	9.0°	9.4	9.0	9.7	10.0	9.8	9.9	9.7
I	22.3°	21.9	21.6	21.4	20.8	20.6	18.8	17.6

N, number of cores per flow.

ξ, angular standard deviation factor (radians), $=(2N - 2\sum_{i=1}^{N} \cos \delta_i)^{1/2}/(N-1)^{1/2}$, where δ_i is angle between individual direction and mean direction.

D, declination of flow average remanent magnetization.

I, inclination of flow average remanent magnetization.

The NRM values cited here are based on repeat measurements and thus differ slightly from those cited in Table 6.

is that the demagnetization process adds a random component of magnetization which varies from one experiment to the next and which becomes relatively greater as the demagnetizing field becomes larger and the amount of remaining NRM smaller.

From the experiments summarized in Table 4, there appears to be no single value of the demagnetizing field that would substantially reduce angular scatter of directions in all flows. For flows like Kahuku unit 114, a complete set of experiments would permit a reduction in angular scatter, but, in view of the apparent rarity of flows of this type, the over-all improvement in precision is not worth the excessive number of experiments required. In subsequent analyses, therefore, the values of NRM without partial demagnetization will be used.

Within-flow analyses. The oriented cores collected for this study were sufficiently long to permit preparation and measurement of more than one specimen from each core. In our earlier

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TABLE 5. Results of Two-Level Statistical Analyses for Within-Core and Between-Core Variations for Selected Flows

Volcanic Series	Unit	B	N	$\hat{\omega}$	$\hat{\beta}$	k_c	$\Delta\alpha_{95}$
Puna	132	9	18	2500	93	85	0.72°
Kahuku	122	8	13	2841	451	394	0 35
Kahuku	114	7	14	1023	297	291	0.26
Kahuku	123	7	14	1085	568	495	0.22

B, number of cores per flow.
N, total number of specimens per flow.
$\hat{\omega}$, within-core precision parameter.
$\hat{\beta}$, between-core precision parameter.
k_c, between-core precision parameter based on measurements of one specimen per core [*Fisher*, 1953].
$\Delta\alpha_{95}$, difference in α_{95} determined from two-level analysis and single-specimen data.

investigation of the historic flows from Hawaii [*Doell and Cox*, 1963], we found that little additional accuracy could be gained by measuring more than one specimen from each core. To similarly determine the increase in accuracy that might be obtained from the present data, we measured the NRM of two specimens from each core collected from four flows and applied a two-level statistical analysis [*Watson and Irving*, 1957]. The within-core scatter is much less than that between cores (Table 5). As a consequence, the gain in accuracy achieved by measuring more than one specimen from each core is scarcely worth doubling the number of measurements. Therefore, in subsequent analyses measurements on one specimen from each core were used.

The average direction of magnetization of each flow is shown in Figures 8 through 12, and is listed in Table 6. All data are based on measurements of the NRM of one specimen from each core. It is important to note that α_{95} (Table 6) applies only to the NRM averaged over that part of the flow from which the cores were collected; it is not a confidence limit about an ancient direction of the earth's main magnetic field. Even given perfect paleomagnetic

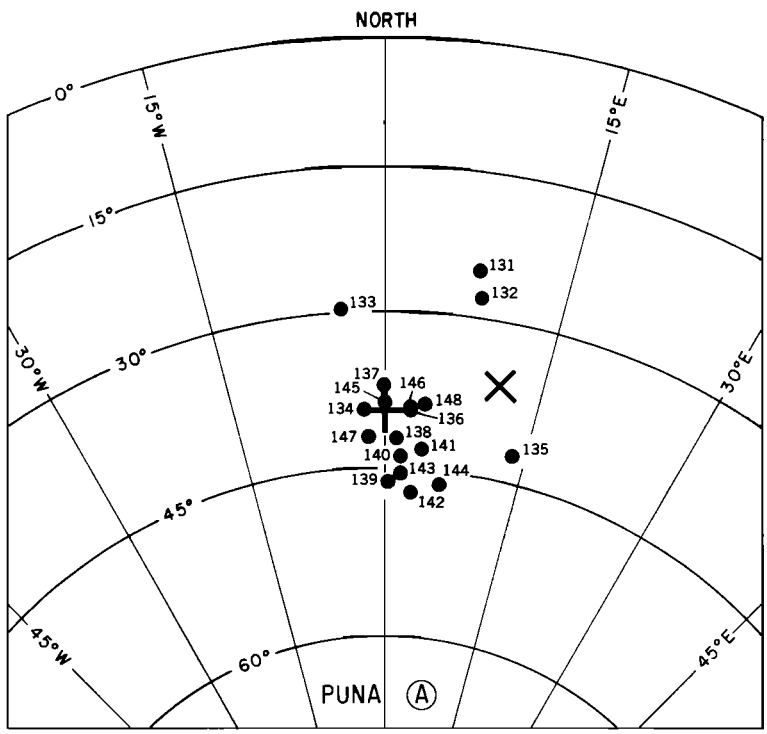


Fig. 8. Directions of NRM for lavas of the Puna volcanic series. ● Lower hemisphere; ○ upper hemisphere; + 'single sample' directions, lower hemisphere; + axial dipole field direction; × inclined dipole field direction. Flows are numbered consecutively from bottom to top. Lambert equal-area projection.

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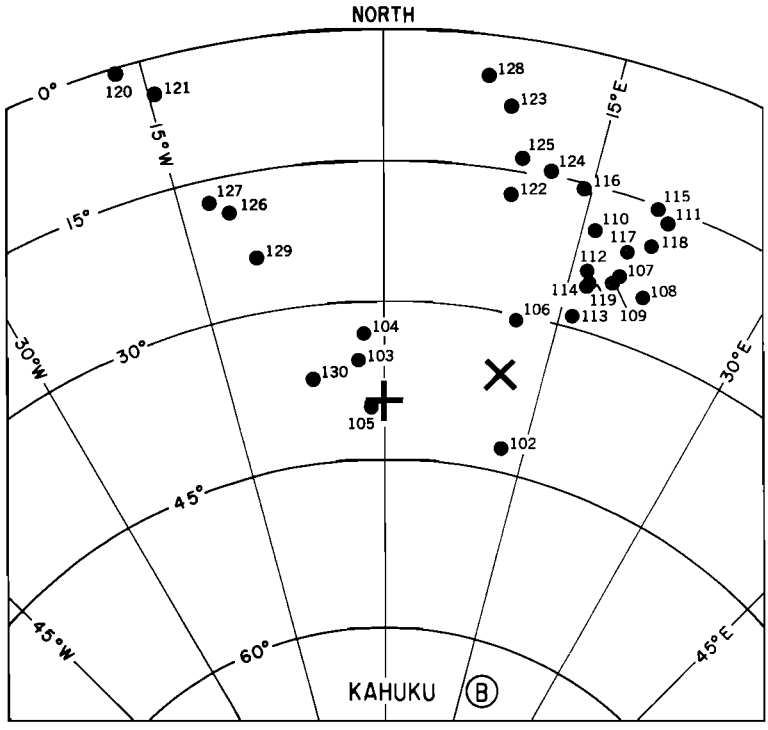


Fig. 9. Directions of NRM for lavas of the Kahuku volcanic series. (See caption for Figure 8.)

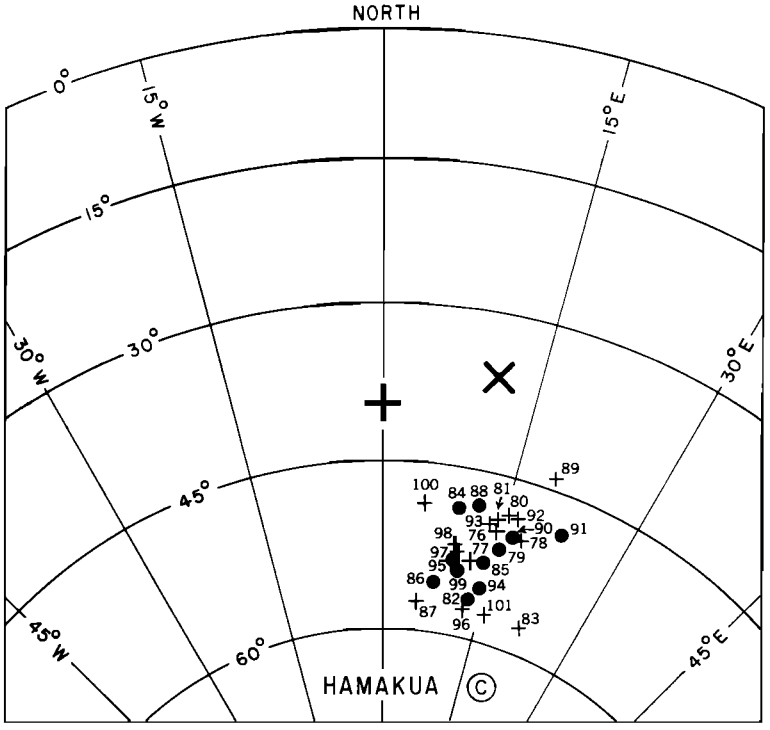


Fig. 10. Directions of NRM for lavas of the Hamakua volcanic series. (See caption for Figure 8.)

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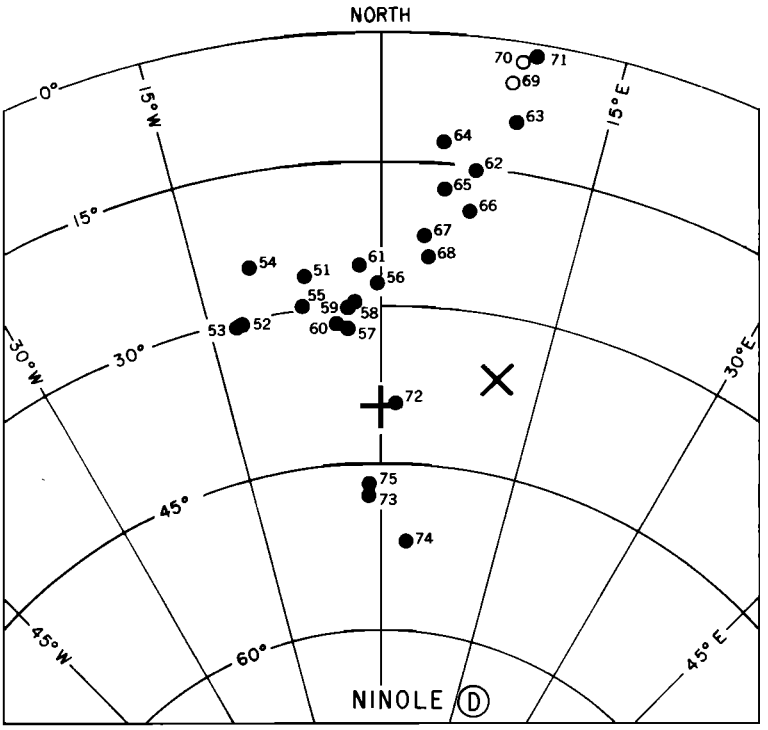


Fig. 11. Directions of NRM for lavas of the Ninole volcanic series.
(See caption for Figure 8.)

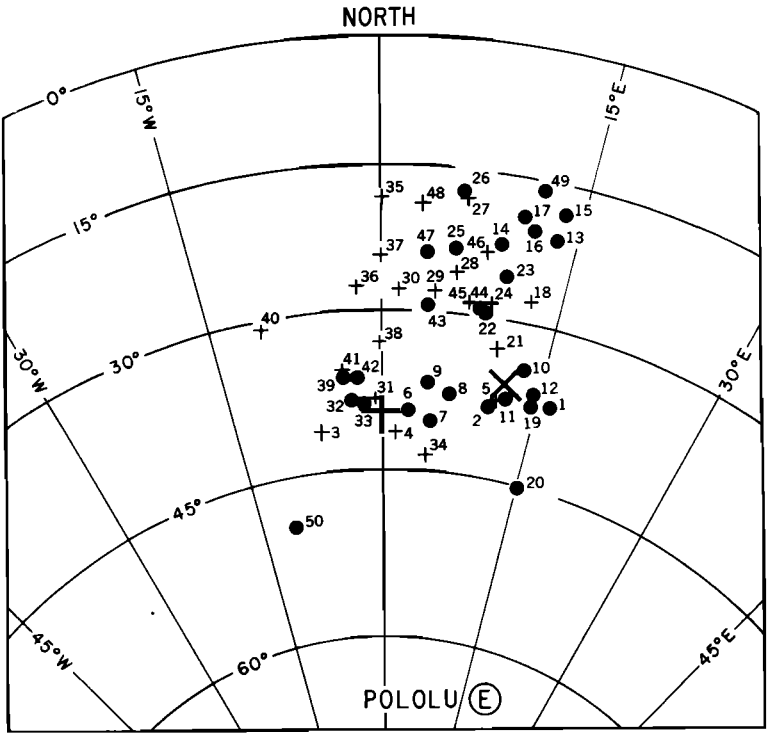


Fig. 12. Directions of NRM for lavas of the Pololu volcanic series.
(See caption for Figure 8.)

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TABLE 6. Flow-Average NRM Data from Hawaiian Lavas

Unit No.	B, m	N	C, m	T, m	$\overline{\Delta D}$ ± s.d., deg	$\bar{J} \times 10^3$ ± s.d.	Decl. of $\bar{J}$, deg	Incl. of $\bar{J}$, deg	R	k_c	ξ_c, deg	α_{95}, deg	γ_{95}, deg
					(a) Puna Volcanic Series								
148	1	5	38	2 4	3.6W ±1.8	10.48 ±4.82	3.86	+38 97	4 95550	89.89	8 55	8 11	8 38
147	6	6	18	3.0	0.5W ±1.9	6.73 ±2.56	358 23	42.06	5.97647	212 54	5.56	4.61	5 07
146	12	5	7	0.9	4.4W ±1.5	9.05 ±1.03	2 58	39.39	4 98612	288 28	4.77	4.51	4 98
145	20	5	20	1.2	0.1W ±2.9	3.05 ±0.44	0 01	38 91	4 95906	97.70	8 20	7.78	8 07
144	47	7	42	1.8	4.4W ±3.1	9.90 ±2.28	6 22	46.32	6.99056	635 36	3 21	2 40	3 20
143	48	8	44	1 5	2.5W ±3.0	11.72 ±2.71	1.80	45 38	7.92931	99 03	8 14	5 59	5.97
142	61	5	17	1 5	1.1E ±3.6	12.11 ±8.68	3.02	47.25	4 97848	185 84	5 94	5 63	6 02
141	67	6	11	0 9(+)	0.5E ±5.3	13.24 ±6.94	3.99	43 30	5 96318	135.78	6.96	5.77	6 14
140	72	7	35	0 6	1.3E ±2.5	7.07 ±1.71	1.79	43 83	6.95752	141.26	6.82	5.10	5 52
139	81	7	21	0 9	3.2E ±3.8	10.47 ±3.53	0 32	46 25	6 96875	191 99	5.85	4 37	4.85
138	88	7	23	1 8	0.1W ±3.8	5.89 ±1.93	1.12	42 16	6 97083	205 68	5 65	4 22	4 72
137	92	6	8	1 2	3.6E ±5.7	10.34 ±1.07	359 96	37.44	5 98637	366 71	4.23	3.50	4 08
136	100	8	48	0 3	7.9E ±1.4	3.15 ±1.16	2 57	39.38	7 81585	38.01	12 44	9 10	9.35
135	112	6	34	2 7	2.7E ±7.3	7.19 ±4.75	13 68	42 52	5 97918	240.11	5 23	4 33	4.82
134	116	8	44	0 9	5.2E ±4.1	11.30 ±4.09	357 91	39 47	7.96235	185 92	5 94	4 07	4.58
133	119	4	5	3 7	2.6E ±2.8	8.67 ±0.07	356 22	29 80	3.99913	3452 24	1 38	1 56	2 62
132	127(+)	9	39	2.4	2.4E ±4.0	8.50 ±2.99	8 03	27 95	8.90604	85.14	8.78	5.61	5 98
131	131(+)	6	32	1 2	3.2E ±3.5	9.07 ±1.71	7 77	25 29	5 98204	278.35	4 85	4 02	4.54
					(b) Kahuku Volcanic Series								
130	2	7	23	3.1	0.9E ±0.8	2.72 ±0.44	353 26	+37 01	6.96610	177 01	6.09	4.55	5 01
129	4	5	20	1 5	1.1W ±2.2	3.24 ±0.71	350 00	24 34	4.97437	156 09	6 48	6 14	6.48
128	10	5	23	1.8	0.2E ±1.3	2.75 ±0.26	6 83	5 10	4 97232	144 52	6 74	6.39	6 73
127	12	5	19	1 2	0.4W ±1.8	3.08 ±0.85	347 28	17 87	4.99583	959 35	2 62	2 47	3 25
126	13	5	16	0 9	0.1W ±1.9	2.32 ±1.29	348.50	19 19	4 99348	613 55	3 27	3 09	3.75
125	16	5	17	1.1	0.5E ±0.4	1.77 ±0.69	9 82	13 97	4.99212	507 74	3 60	3.40	4 00
124	18	5	23	1 4	1.9E ±1.3	2.31 ±1.22	11 91	14 58	4 98822	339 53	4 39	4 16	4 67
123	19	5	35	0 3	1.3W ±4.0	7.14 ±2.16	8 52	8 39	4 97320	149 28	6 64	6 28	6 63
122	22	8	64	3 7	0.8W ±1.2	2.43 ±0.58	9 43	17 83	7 98222	393 65	4.08	2 80	3 50
121	24	5	46	3.4	0.3W ±1.3	1.31 ±0.49	345 41	4.49	4 98571	279 85	4 84	4 58	5 05
120	27	5	47	0 3	0.2E ±0.4	0.82 ±0.20	343 54	1 00	4 98608	287 39	4.78	4 52	4.98
119	31	5	17	1 7	0.1E ±0.5	1.08 ±0.19	16.70	25 35	4 93199	58 81	10 58	10 06	10 28
118	33	5	18	1 2	0.3E ±0.4	1.18 ±0.40	20 36	19.54	4 94912	78 61	9.15	8.68	8 93
117	41	7	75	3 8	1.3E ±2.0	4.22 ±3.28	18 80	20 99	6 95889	145 95	6.71	5 01	5 45
116	45	5	7	0 3	3.2W ±3.7	19.38 ±12.51	14 47	15 91	4 98236	226 72	5 38	5 09	5 50
115	49	5	12	0 5	2.0E ±0.7	6.59 ±0.65	19.91	15 63	4.97684	172 74	6.16	5 84	6 20
114	59	7	59	6 1	0.1W ±1.3	1.96 ±3.00	16 38	25 71	4 97936	290 70	4 70	3 55	4 13
113	65	5	32	2 4	1.0E ±0.8	0.68 ±0.33	16.03	28.91	4 98728	314 41	4 57	4.32	4 81
112	68	5	21	0 9	2.0E ±2.0	5.61 ±1.88	16 11	24 04	4 97604	166 92	6 27	5 94	6 30
111	70	6	30	0 8	2.8E ±2.8	9.52 ±1.78	20 88	16 69	5 97784	225.59	5 39	4 47	4 95
110	70	5	10	0 3	3.0E ±2.7	9.86 ±1.49	15 91	19 85	4.97559	163 85	6 33	6.00	6 37
109	75	5	28	2 0	0.3W ±0.9	1.63 ±0.35	18 43	24 34	4 99611	1027 27	2 53	2 39	3 19
108	76	5	7	0.3	0.4W ±0.4	1.20 ±0.13	20 99	24 80	4 98259	229 75	5 34	5 06	5 49
107	79	5	10	1.1	0.2E ±0.2	0.80 ±0.21	18 79	23 81	4 99648	1136 11	2 41	2 27	3.10
106	84	5	12	0 3	1.4E ±0.6	0.53 ±0.21	11.55	30.65	4 99246	530 60	3 52	3 32	3 93
105	89	5	26	1 8	1.3E ±1.0	1.84 ±0.19	358 91	40 04	4 97606	167 09	6 27	5.94	6.30
104	95	8	92	3.4	1.6W ±1.6	3.94 ±0.87	358 04	33 10	7 97523	282 57	4 82	3 30	3.92
103	98	5	34	1 1	1.1W ±0.8	1.16 ±0.34	357 58	35 62	4 98422	253 43	5 08	4.82	5 26
102	102	5	21	1.8	0.7W ±0.4	1.60 ±0.43	12 68	42.91	4 98944	378 74	4.16	3.94	4.47
					(c) Hamakua Volcanic Series								
101	0	1	...	0.9	3.0E ± ...	6.69 ± ...	16 2	+57.7	...	...	...	...	...
100	2	1	...	...	7.5E ± ..	15.9 ± ...	5.1	48 6	...	...	.	...	...
99	2	5	14	0 6	1.3E ±1.5	7.38 ±2.76	10 80	54 18	4 99276	552 19	3 44	3 26	3 88
98	3	1	...	0 9	0.5E ± ...	13.9 ± ...	9 6	52 0	...	...	...	...	...
97	5	1	...	...	2.0E ± ...	7.97 ± ...	10 2	52 7	...	...	...	...	...
96	6	1	.	...	3.5E ± ...	3.52 ± ...	12 4	57.6	...	. .	..	...	...
95	6	6	36	1.5	4.7E ±1.5	7.70 ±5.22	9 52	53 34	5 96948	163 82	6 33	5 25	5 67
94	8	7	23	1 4	4.6E ±2.6	7.88 ±2.07	14 25	55 52	6.98787	494 70	3 64	2 72	3.44
93	10	1	...	0.9	7.0W ± ...	5.10 ± ...	13.6	49 5	...	...	...	...	...
92	11	1	...	...	8.5W ± ...	5.26 ± ...	16.7	48 4	...	...	...	...	...
91	12	5	7	1.4	0.4W ±8.3	5.87 ±1.54	22.33	48 61	4.94973	79 56	9 08	8 63	8.88
90	13	6	18	1.0(+)	4.2W ±1.8	5.51 ±1.61	16.66	50.10	5.99207	630.14	3.22	2 67	3 40

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TABLE 6. (Continued)

Unit No.	B, m	N	C, m	T, m	$\overline{\Delta D}$ ± s.d., deg	$\bar{J} \times 10^3$ ± s.d.	Decl. of $\bar{J}$, deg	Incl. of $\bar{J}$, deg	R	k_a	ξ_a, deg	α_{95}, deg	γ_{95}, deg
89	15	1	...	...	5.0W ± ..	10.5 ±...	19.3	43 9	...	...	...	...	...
88	18	5	9	0 6(+)	6.3E ±1.1	6.00 ±1.42	11 78	48 01	4 97515	160 98	6 21	6.05	6 42
87	22	1	...	...	3.0E ±...	2.87 ±...	5 1	57 5	...	..	...	...	...
86	24	4*	38	...	4.6E ±3.3	5.85 ±0.79	7 50	55 63	3.98626	218 35	5 48	6 23	6 58
85	27	4	30	1 2	8.9E ±0.5	7.55 ±1.03	13.74	53.12	3 99137	347 54	4 34	4 94	5 37
84	29	5	33	0.9	4.2E ±3.2	7.91 ±0.64	9 10	48 78	4 95330	85 65	8 76	8.31	8.67
83	31	1	...	0.3	2.0W ±...	10.6 ±...	21 7	57 8	...	...	...	...	..
82	31	5	9	0 6	2.3E ±1.6	6.56 ±2.74	12 91	56 91	4 97610	167 34	6 21	5.93	6 30
81	32	1	...	...	0.5W ±...	6.12 ±...	14 3	49 0	...	...	...	...	...
80	34	1	...	0 6	3.0E ±...	6.46 ±...	15 5	48 5	...	...	...	...	...
79	35	6	9	0 9	3.2E ±0.5	7.25 ±2.36	15 45	51.70	5 99428	317 28	4 55	3.77	4 32
78	37	1	...	...	2.0E ±...	5.48 ±...	17 7	50.1	...	...	...	...	...
77	37	1	...	...	1.5W ±...	2.69 ±...	12 1	53 3	...	...	...	...	...
76	38	1	...	...	4.5W ±...	5.90 ±...	14 6	50 0	...	...	...	...	...
						(*d*) Ninole Volcanic Series							
75	1	10	61	2.4	1.9W ±1.6	3.10 ±1.42	358 81	+46 85	9 97719	394 63	4 07	2 43	3.22
74	3	9	41	0.9	1.6W ±0.8	3.97 ±1.11	3 18	52 05	8 95900	195.12	5 80	3 69	4 25
73	5	6	28	4 3	1.6W ±1.2	4.28 ±1.25	358 46	47 90	5 98390	310 49	4 59	3 81	4 37
72	9	6	23	3.0	0.7W ±1.5	6.61 ±1.92	1 53	39 19	5 98411	314 73	4 57	3 78	4 33
71	12	5	49	1 5	0.8W ±0.7	5.20 ±0.58	9 72	1 91	4.97715	175 08	6 12	5 80	6.17
70	15	5	22	5 5(+)	3.1W ±2 2	4.03 ±1.81	8 90	−2 78	4 97348	150 82	6 60	6 25	6 60
69	19	5	11	2 1	0.1E ±2.3	4.00 ±0.96	8 47	−5 17	4 97361	151 58	6 58	6 23	6 58
68	20	5	8	1.2	1.1W ±1.7	3 91 ±1.13	3 76	+24 95	4 98763	323 36	4 50	4 26	4 46
67	22	5	8	2 1(+)	3.0W ±0.7	6.10 ±0.74	3 32	22 82	4 95922	98 08	8 17	7 76	8 05
66	23(+)	5	7	1 2	0.7W ±2.2	5.33 ±1.67	6 62	19 88	4 97288	147 50	6 67	6 32	6 67
65	24(+)	5	9	1 4	1.3E ±1.8	6.28 ±1.21	4 66	17 87	4 95014	80 23	9 05	8 59	8 83
64	26(+)	5	9	1 8	1.7E ±1.3	5.74 ±0.59	4 33	12 66	4.99006	402 53	4 04	3 82	4 37
63	28(+)	6	10	2 1	0.3E ±2.0	5.64 ±1.72	9 00	9 74	5 98425	317.49	4 55	3 77	4 32
62	31(+)	6	12	3.7	0.8E ±1.3	5.46 ±1.17	6 63	15 42	5 98708	386 90	4 12	3 41	4 02
61	34(+)	6	5	2 7	0.6W ±0.9	6.18 ±1.19	358 10	25 82	5.97157	175 84	6 12	5 07	5 49
60	36(+)	5	8	2 1	0.3W ±0.9	4.35 ±0.71	356 10	31 57	4 99684	1266 58	2 28	2.15	3 01
59	38(+)	5	11	1 8(+)	1.6W ±0.7	4.13 ±1.15	357 04	29 98	4 99809	2090 19	1 77	1 67	2 68
58	41(+)	6	6	2 4	0.8W ±0.7	4.60 ±0.88	357.72	29 45	5.99375	800 50	2 87	2 37	3 17
57	43(+)	7	14	2 7	0.1E ±1.3	5.60 ±1.45	356 99	32 08	6 98862	527 03	3 53	2 63	3 38
56	46(+)	7	20	2 4	0.4W ±0.7	3.97 ±1.62	359 65	27 56	6 97228	216 42	5 51	4 11	4 63
55	48(+)	11	32	2 4(+)	0.5W ±1.0	5.56 ±1.52	353.23	29 42	10 98408	628 26	3 23	1 82	2 78
54	51(+)	4	10	2 1	2.1E ±0.4	6.96 ±0.89	349 50	24 92	3 96333	81.82	8 96	10 22	10 43
53	53(+)	5	11	2.4	2.4W ±1.8	6.78 ±0.76	347 37	30 61	4 97892	189 75	5 88	5 57	5 95
52	55(+)	6	23	1.5(+)	0.2E ±1.1	6.01 ±2.08	347 87	30 44	5 98877	445 36	3 84	3 18	3 82
51	56(+)	9	34	1 7	1.4E ±1.12	6.20 ±1.29	353 68	26 49	8 76375	33 86	13.93	8 98	9 23
						(*e*) Pololu Volcanic Series							
50	1	5	5	1.1	0.7W ±0.2	2.25 ±0 52	349 01	+49 55	4 99487	780 26	2.90	2 74	3 46
49	4	5	8	0 6	2.3W ±1.7	2.93 ±0.91	11 83	16 05	4 99367	631.42	3 22	3.05	3 72
48	13	1	...	0 9	3.0W ±...	3.67 ±...	3 1	18 8	...	...	...	. .	. .
47	17	5	10	3 4	0.1E ±1.5	3.55 ±1.02	3 78	23.91	4.99556	900 17	2 70	2 55	3 32
46	20	1	...	0 6	1.0W ±...	3.54 ±...	8 3	23 2	...	...	...	...	...
45	22	1	...	...	2.0W ± ..	4.55 ±...	7 4	28 5	...	...	...	...	...
44	25	5	5	0 8	2.1W ±1.2	4.50 ±2.06	8 47	29 08	4 99556	900 50	2 70	2.55	3 32
43	28	5	7	0 5	1.0E ±1.4	6.87 ±8.72	4 03	29 13	4 96471	113 33	5 38	7 22	7.52
42	30	5	5	0 6	3.5W ±1.3	7.88 ±1.10	357 78	36 32	4 99181	488 63	3.66	3 46	4 06
41	31	1	...	0.5	1.5W ±...	6.00 ±...	356 3	34 5	...	...	...	...	...
40	33	1	...	0 3	9.0W ±...	12.8 ±...	349.6	30 7	...	...	...	...	...
39	35	5	6	1 2	8.9W ±1.4	8.08 ±1.05	356 30	35 20	4 99625	1066.72	2.48	2.34	3 15
38	36	1	...	0.6	7.5W ±...	4.64 ±...	359 8	32.9	...	...	...	...	...
37	37	1	...	0 9	4.0W ±...	6.14 ±...	0 0	24.1	...	...	...	...	...
36	38	1	...	0 6	4.0W ±...	5.16 ±...	357 9	27.2	...	...	...	...	...
35	39	1	...	0.3	3.5W ±...	7.76 ±...	0.1	18.1	...	...	...	...	...
34	41	1	...	0 9	0.5W ±...	5.75 ±...	4 7	43.5	...	...	...	...	...
33	42	5	4	1.2	4.3W ±2.0	5.85 ±0.35	358 09	38 89	4.99536	862.40	2 76	2 61	3 36
32	43	5	4	0 5	7.3W ±1.1	7.84 ±1.64	356.96	38.55	4 97741	177 04	6 08	5 77	6.14
31	48	1	...	0 3	8.0W ±...	4.95 ±...	359.3	38.4	...	...	...	...	...
30	49	1	...	0.3	4.0W ±...	2.93 ±...	1.6	27.8	...	...	...	...	...

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TABLE 6. (Continued)

Unit No.	B, m	N	C, m	T, m	$\overline{\Delta D}$ ± s.d., deg	$\overline{J} \times 10^3$ ± s.d.	Decl. of $\overline{J}$, deg	Incl. of $\overline{J}$, deg	R	k_c	ξ_c, deg	α_{95}, deg	γ_{95}, deg
29	50	1	...	0.3	1.0W ± ...	4.71 ± ...	4.5	27.8	...	...	...	..	. .
28	51	1	...	0.3	0.5E ± ...	2.28 ± ...	6.2	25.8	...	...	...	...	...
27	53	1	...	0.3	2.5W ± ...	2.12 ± ...	6.5	18 0	..	...	...	...	...
26	54	6	11	...	1.7W ±0.6	4.26 ±1.22	6 02	17.36	5.96384	138 28	6 89	5 72	6.10
25	59	5	12	3.4	0.8E ±0.6	2.96 ±0.30	5.92	23 29	4 99556	901 27	2 70	2.55	3.32
24	61	1	...	0.3	2.0W ± ...	5.01 ± ...	9 4	28 3	. .	...	. .	...	...
23	62	5	5	1.2	0.9W ±1.00	4.37 ±0.61	10.14	25 45	4 99017	406.75	4 02	3.80	4.35
22	65	5	11	4.6	2.0W ±0.5	5.00 ±0.92	8.83	29 69	4 99749	1593.31	2 03	1.92	2.85
21	68	1	...	...	2.0E ± ...	4.67 ± ...	10.5	32.8	...	...	...	...	...
20	70	5	5	1.1	2.0W ±1.0	4.66 ±0.81	15.05	45.35	4 98750	319.98	4 53	4.28	4 78
19	74	5	6	2.1	0.1W ±1.5	5.11 ±1.03	14.45	37.76	4 99387	652.18	3 17	3 00	3 67
18	78	1	...	...	0.5W ± ...	1.16 ± ...	12.6	27.7	...	...	...	...	...
17	81	5	6	...	3.0E ±1.9	5.77 ±1.35	10.78	19.16	4 99851	2684.38	1.56	1 48	2.58
16	84	5	5	0.6	0.2W ±1.3	3.52 ±0.36	11 76	20.57	4 99746	1572.51	2.04	1 93	2.87
15	88	5	4	1.2	0.8E ±2.0	4.63 ±2.48	13.71	18.13	4.98540	273.99	4.90	4.63	5.10
14	92	6	...	2.4	2.3E ±1.2	3.98 ±1.54	9.31	22 30	5 99185	613.12	3.27	2.71	3.44
13	95	5	7	1 8	0.3W ±0.9	4.35 ±2.02	13.45	21 00	4.98745	318.67	4.53	4 29	4.78
12	96	5	6	...	1.2E ±1.4	6.15 ±2.48	14.56	36 63	4.99715	1401.05	2.16	2.04	2.93
11	104	5	13	1.5	0.3W ±0.9	9.75 ±3.64	11 99	37 47	4 99128	458.48	3.78	3.58	4.17
10	107	5	12	3.0	2.4W ±1.6	5.22 ±0.8	13.10	34.42	4.99491	785.92	2 89	2 73	3.46
9	116	5	5	2.1	1.3W ±1.0	7.11 ±1.25	4.35	36 71	4 99062	426.32	3.92	3 71	4.28
8	119	5	10	3.0	2.5W ±1.3	6.72 ±1.77	6.58	37.71	4 99602	1004.47	2.53	2 42	3.21
7	123	5	8	3 7	4.7W ±0.8	6.43 ±2.69	4 79	40.40	4.99019	407.54	4 02	3.79	4.33
6	125	5	10	1.2	4.9W ±1.0	7.94 ±2.35	2.57	39.30	4 99614	1037.05	2 52	2 38	3.19
5	126	1	...	...	5.0E ± ...	3.68 ± ...	11.2	37.8	...	...	...	...	...
4	128	1	...	...	2.5W ± ...	3.13 ± ...	1.4	41.5	...	...	...	...	...
3	130	1	...	...	4.5W ± ...	2.87 ± ...	353 8	41.1	...	...	...	...	...
2	133	5	12	...	3.7W ±0.5	1.88 ±0.99	10.31	38.49	4.95241	84.06	8.83	8.39	8.65
1	138	5	14	0.9	1.8W ±0.6	1.69 ±0.31	16.33	37.29	4.99246	530 23	3.51	3.33	3.94

B, distance below top.
C, core spacing.
T, average flow thickness.
N, number of sample cores.
$\overline{\Delta D}$, average declination error at core locations.
$\overline{J}$, intensity of NRM (emu/cc).
R, vector sum of N unit vectors.
k_c, intraflow precision parameter $(N - 1)/(N - R)$ [*Fisher*, 1953].
ξ_c, intraflow angular standard deviation factor as defined in Table 4.
α_{95} 95 per cent confidence cone about average direction of magnetization.
γ_{95} 95 per cent confidence cone about ancient geomagnetic field direction.
* one core omitted.

technique, the most we can obtain in a paleomagnetic investigation is the average direction of the geomagnetic field over a rather limited space in which local anomalies may be produced by sources in the nearby volcanic terrain. These anomalies add to the main field originating in the earth's core, which is the field of primary interest in paleomagnetism, and thus constitute a 'noise signal.'

In our earlier study of historic flows, we estimated that in Hawaii these anomalies produce an average deflection of 1¼° over the region routinely sampled [*Doell and Cox*, 1963]. The anomalies which existed at the time a flow originally cooled cannot in general be reconstructed, so that the deviations produced by the anomalies are best treated as additional random errors. Ninety-five per cent confidence limits which take into account this effect are given in Table 6, where γ_{95}, the radius of the cone of confidence, is given by the expression

$$\cos \gamma_{95} \approx \cos \alpha_{95} + (1/k_L) \ln (.05)$$

where k_L is the precision parameter corresponding to the local anomaly field [*Doell and Cox*, 1963]. A value of $k_L = 4400$ was used in calculating values of γ_{95}. At the time these flows erupted, the direction of the main field in Ha-

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waii was within γ_{95} degrees of the directions defined by the declination and inclination values indicated in Table 6. These we take as the primary paleomagnetic data.

Analysis of flow averages. The statistical analysis most commonly used in paleomagnetism is that of *Fisher* [1953], which is based on the following three assumptions about the statistical sample. (1) The sequence in time is not taken into account, but rather it is assumed that each datum is drawn randomly from a population of unit vectors describing successive directions of the geomagnetic field. (2) The population of unit vectors is assumed to have a radial density distribution describable by the function

$$(\kappa/4\pi \sinh \kappa)e^{\kappa \cos \alpha}$$

where κ, a parameter characterizing the population, is an inverse measure of angular dispersion and α is the angular distance from the direction of greatest density. (3) The population of vectors is assumed to be distributed with azimuthal symmetry about a mean direction.

Despite the almost universal use of this statistical method in paleomagnetism, only for a few sets of data has the validity of these three assumptions been checked against observed populations [*Watson and Irving,* 1957]. Because a rather large population of data points is needed for a definitive test, the present set of observations from Hawaii may be useful for this purpose. Assumption (1) implies that there should be no correlation between successive directions of magnetization in a series, so that the angular distance between directions derived from adjacent flows should, on the average, be no less than that between widely separated flows. Simple inspection of Figures 8 to 12 is sufficient to indicate that this condition is not satisfied in the Hawaiian data. Assumptions (2) and (3) have been tested by the methods of *Watson and Irving* [1957] which were designed to estimate the significance of the departure of an observed population from the model of Fisher. These tests have been applied both to the observed directions and to virtual geomagnetic poles (VGP) [*Cox and Doell,* 1960] calculated from these directions, and the results are given in Table 7. As would be anticipated from inspection of the figures, departures of individual series from the theoretical distribution of Fisher is significant at rather high probability levels. (Too few data are available from the Hamakua volcanic series for a useful test.) Less obvious and more interesting is the result that the population of data points of all 112 flows from which multiple cores were collected is considerably different from the Fisher distribution both in azimuthal symmetry (99.9% significance) and in radial dependence (99.5% significance).

Despite the fact that none of the above conditions is satisfied, it is of interest to examine the consequences of formally applying Fisher's statistical method to these data (Table 7). The results are also presented graphically in Figure 13. These calculations show the interesting result that the average direction of the geomagnetic field during the extrusion of the Hawaiian lavas was significantly different from that of an axial dipole field at the 95% confidence level and that the mean virtual geomagnetic pole is significantly different from the present rotation axis. These conclusions would thus appear to be in disagreement with other paleomagnetic studies of rocks of similar age [*Cox and Doell,* 1960]. The result of ignoring departures of the observed data from the assumed population of Fisher could attach a false significance to the observed departure from the axial dipole field, and this, in turn, might easily be translated into an erroneous probability statement about the drift of Hawaii relative to the large land masses. On the other hand, the departures could be real; but it is clear that their significance cannot be determined by Fisher's method.

The reasons for the original departure of the Hawaiian paleomagnetic data from the model of Fisher are simple to discern. At any locality the geomagnetic field vector undergoes continuous rather than discontinuous changes in orientation. Thus any set of data points distributed over a sufficiently short time interval cannot possibly approximate the statistical model of Fisher. This model can only be satisfied when the time interval between successive data points is longer than all of the periods involved in changes of the orientation of the geomagnetic field vector. Although little or nothing is known about the longer periods associated with geomagnetic secular variations, it is clear that on Hawaii either the rates of secular variation were too slow or the rates of volcanism were too rapid to permit an approximation of the data by Fisher's model.

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TABLE 7a. Statistical Analyses of Flow-Average Directions of NRM

Volcanic Series	N	$\hat{\chi}$	$\hat{\psi}$	k	R	α_{95}, deg	ξ, deg	D, deg	I, deg
Puna	18	70	75	130	17.87	3.0	7.1	2.8	+39.8
Kahuku	29	96.5	85	28	28.01	5.1	15.3	7.9	+22.2
Hamakua	11			402	10.98	2.3	4.0	13.2	+52.4
Ninole	25	99.9	96.5	28	24.13	5.6	15.3	0.5	+25.0
Pololu	29	95	97.5	59	28.52	3.5	10.6	7.5	+31.7
All flows	112	99.9	99.5	27	107.92	2.6	15.6	5.7	+31.1

TABLE 7b. Statistical Analyses of Flow-Average Virtual Geomagnetic Poles

Volcanic Series	N	$\hat{\chi}$	$\hat{\psi}$	k	R	α_{95}, deg	σ, deg	Lat., deg	Long., deg
Puna	18	35	75	193	17.91	2.5	5.8	+85.7	240.6
Kahuku	29	99.5	93	37	28.25	4.4	13.3	+79.6	336.2
Hamakua	11			313	10.97	2.6	4.6	+72.4	243.8
Ninole	25	99.9	99 5	57	24.58	3.9	10.7	+84.6	21.7
Pololu	29	70	96.5	90	28.69	2.8	8.5	+82.6	313.5
All flows	112	99	99.5	46	109.56	2.0	12.0	+84.2	312.9

N, total number of flows in each section (only those flows with four or more cores were included in these analyses).

$\hat{\chi}$, $\hat{\psi}$, probabilities that distributions are non-Fisherian; $\hat{\chi}$, azimuthal dependence, $\hat{\psi}$, radial dependence [*Watson and Irving*, 1957].

k, between-flow precision parameter estimate [*Fisher*, 1953].

R, vector sum of N unit vectors.

α_{95}, 95% confidence angle about direction of R.

ξ, angular standard deviation factor for mean flow directions.

σ, angular standard deviation factor for virtual geomagnetic poles.

D, I, declination and inclination of mean flow direction.

Lat., Long., latitude and longitude of virtual geomagnetic poles.

TABLE 7c. Two-Level Analyses of Flow-Average Directions

Volcanic Series	N	S	$\hat{\omega}$	$\hat{\beta}$	α_{95}, deg
Puna	18	115	132	154	2.9
Kahuku	29	158	214	30	4 8
Hamakua	11	58	194	699	2.1
Ninole	25	154	174	28	5 1
Pololu	29	147	388	60	3.4
All flows	112	632	198	29	2.4

N, S, total number of flows and cores in each section.

$\hat{\omega}$, $\hat{\beta}$, within-flow and between-flow precision parameter estimates [*Watson and Irving*, 1957].

α_{95}, 95% confidence angle about mean, calculated from $\hat{\omega}$ and $\hat{\beta}$.

The angular standard deviation factor ξ for sets of mean flow directions is given in Table 7, along with the corresponding precision parameter estimate $k = 2/\xi^2$ for each set. The angular dispersion observed between flow-averaged NRM directions is of dual origin. The larger part is the true angular standard deviation ξ_F between flow-average directions; this is the value of ξ that would be observed if each flow direction were known with perfect accuracy. Superimposed on this is an additional angular dispersion resulting from deviations of the calculated from the true flow-average direction. If ξ_c is the average within-flow angular deviation factor and N is the number of samples in each flow, the observed between-flow angular deviation factor ξ is given by

$$\xi = (\xi_F^2 + \xi_c^2/N)^{1/2}$$

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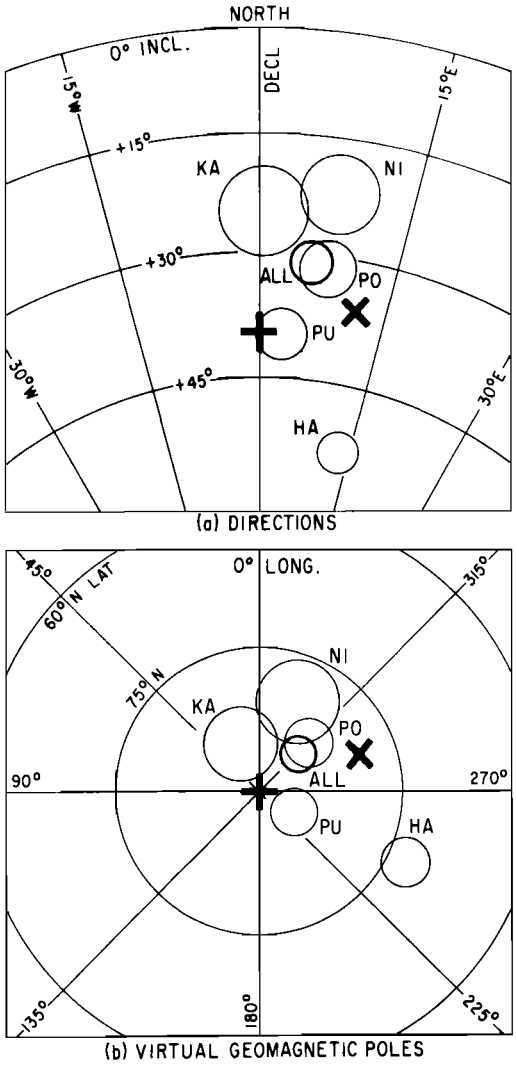


Fig. 13. (*a*) Section-average directions of NRM with 95% confidence limits. (*b*) Section-average virtual geomagnetic poles with 95% confidence limits. Pu, Puna volcanic series; Ka, Kahuku volcanic series; Ha, Hamakua volcanic series; Ni, Ninole volcanic series; Po, Pololu volcanic series; All, all data; ×, inclined dipole field direction and pole; +, axial dipole field direction and pole. Lambert equal-area projection.

Using an average value of 2.4° for $\xi_c/N^{1/2}$, we find a value of $\xi_F = 15.4°$, corresponding to the observed value $\xi = 15.6°$ for all flows combined. The correction is small because the uncertainty in the measurement of flow directions is much less than the angular dispersion between flows.

In an alternative method for evaluating the true angular dispersion between flows, a two-level statistical analysis is used [*Watson and Irving*, 1957]. Although this method in which the same intraflow angular dispersion is assumed for all flows is not strictly applicable to the present data, it has been applied as a check on the method just described for evaluating ξ_F. From Table 7*c*, $\hat{\beta}$, the two-level value for the true between-flow precision parameter is seen to be equal to 29, corresponding to a ξ_F value of 15.1°. The small difference between this and the previously found value of 15.4° indicates that calculations of angular dispersion are not particularly sensitive to the statistical approach used.

Comparing Tables 7*a* and 7*c*, we see that the two-level analysis produces a smaller value for the calculated 95% confidence limits. However, these lower confidence limits are meaningless because the two-level analysis, like that of Fisher, is based on Fisher's model of population distribution, which is not applicable to the present population.

Intensities of NRM. The intensities of the NRM of the Hawaiian lavas are extremely inhomogeneous, both on the level of variation between different cores from one flow and between different flows from one series. Some typical values are shown in Figure 14. Within individual flows the intensity commonly varies by a factor of 2 or 3 from place to place, which is in striking contrast with the homogeneity in directions of magnetization. Within individual volcanic series, average intensities for flows commonly vary by a factor of 3 to 4.

Magnetic anomalies over volcanic terrain are controlled by the remanent magnetization of the flows, which generally is at least 10 times greater than the induced magnetization. The great inhomogeneity of magnetic intensities in lava flows poses great difficulties in relating magnetic anomaly fields to directions of remanent magnetization.

Geological Age of Hawaii

In Hawaii, as in other volcanic islands, attempts to determine geologic ages accurately have been thwarted by the absence of fossils or other evidence useful for stratigraphic correlation. A Pliocene age for the oldest lavas exposed on Hawaii has commonly been quoted [*Stearns*

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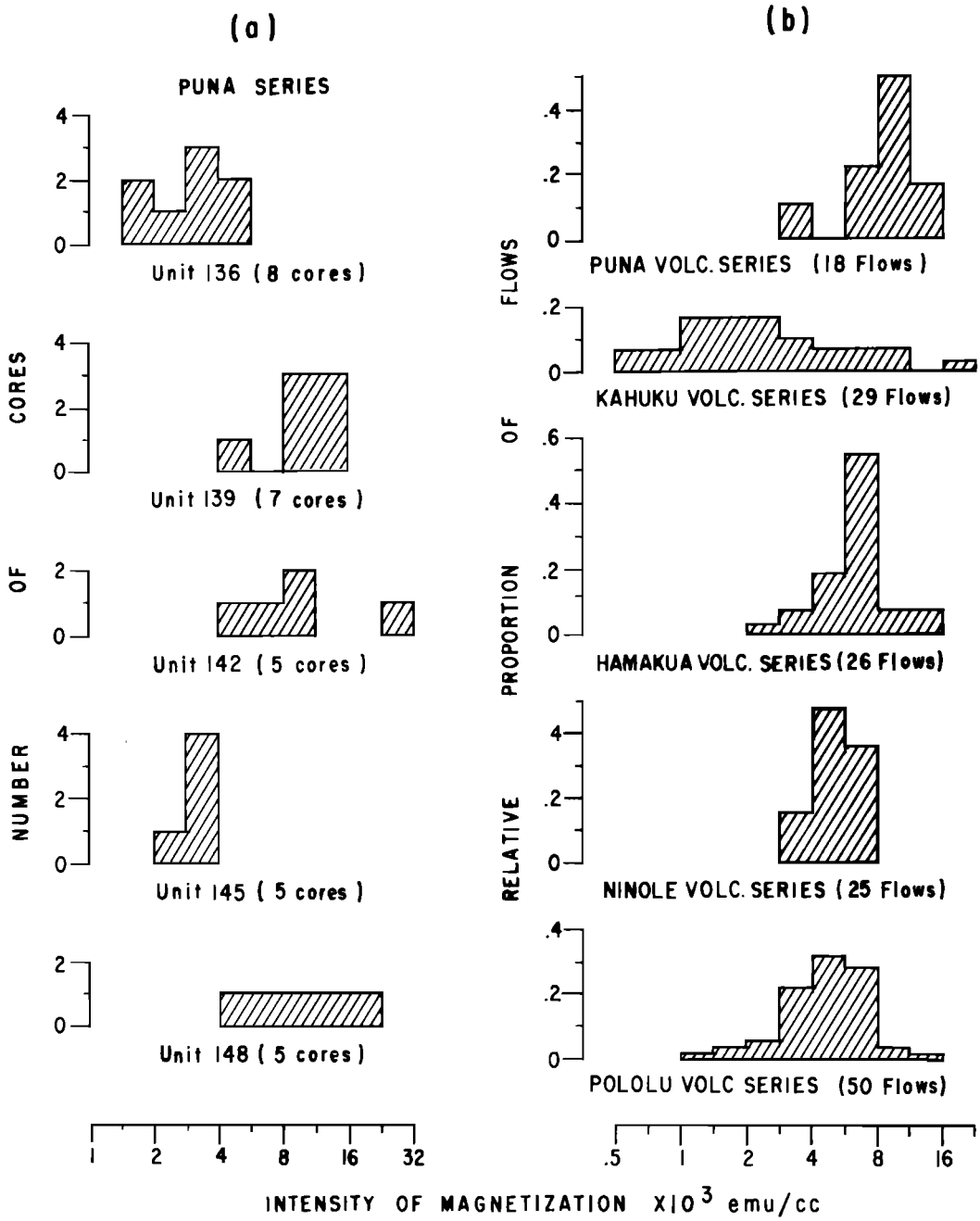


Fig. 14. Histograms of intensities of NRM measured in the Hawaiian lavas; (*a*) within selected flows from the Puna volcanic series, (*b*) flow-average data for all series.

and MacDonald, 1946; *MacDonald and Davis,* 1956], and in a general way this seems reasonable in view of the deep, highly eroded canyons on the northeast coast of the island. Moreover, at one time there appeared to be evidence of four glacial advances on Mauna Kea [*Wentworth and Powers,* 1941]; on re-evaluation of the evidence, however, it now appears that there is strong evidence for only one major advance, presumably Wisconsin in age [*Stearns,* 1945].

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Another reason for assigning a Pliocene age was that the great valleys on the east side of Kohala Mountain were assumed to have been cut before a submergence of Hawaii which is correlated in time with the well-recognized Lualualei submergence of the island of Oahu. The Lualualei submergence, in turn, was ascribed in a very general way to the Pliocene or Pleistocene [*Stearns,* 1935], although fossil control was lacking. Thus, although relative ages of the different volcanic series on Hawaii are well established, the age of the oldest exposed rocks has long remained unknown.

In our preliminary report on the paleomagnetism of the Hawaiian lavas [*Doell and Cox,* 1961], we indicated that all the lavas now exposed on Hawaii are probably younger than early Pleistocene because their directions of magnetization are all normal; that is, they are within several tens of degrees of the present magnetic field in Hawaii. The reason for this stratigraphic assignment is that the earth's magnetic field underwent a worldwide reversal in polarity during the Pleistocene. Wherever a fairly complete section of Pleistocene rocks has been studied paleomagnetically, as in Iceland, for example, two magnetic groups may be recognized—an upper one magnetized normally and a lower one magnetized reversely, that is, in a direction approximately 180° from that of the present field. The boundary between these two groups is thought to occur somewhere near the middle of the Pleistocene, and the radiometrically determined age of this boundary is 1.0 ± 0.2 m.y. [*Cox et al.,* 1964]. An age younger than early Pleistocene can be assigned to the oldest rocks on Hawaii on the basis of these paleomagnetic results unless, as is unlikely, volcanic activity on Hawaii stopped during times when the earth's field was reversed or unless, as is also unlikely, reversely magnetized rocks are presently exposed on Hawaii but were inadvertently missed during the course of the sampling.

The more complete results presented here serve to confirm the earlier interpretation. The 735 samples collected on the island have normal magnetizations, and laboratory tests conducted on these samples all indicate that this magnetization is TRM acquired in the direction of the geomagnetic field at the time the samples originally cooled.

This interpretation of a post-early Pleistocene age for the oldest exposed rocks on the island of Hawaii has recently been confirmed by K-Ar dating. Dates from our units 51 and 54 in the Ninole section given by *Evernden et al.* [1964] are 0.53 and 0.07 m.y. in ascending stratigraphic sequence, and dates from our units 1 and 6 in the Pololu section are 0.00 and 0.43 m.y., in reverse stratigraphic sequence. *McDougall* [1964] has independently analyzed samples from the same Pololu section and reports no significant radiogenic argon. *Evernden et al.* [1964] calculate an average age of 0.25 m.y. from their four data, but in view of the stratigraphic inconsistency in their data and the problem of low potassium and high atmospheric contamination reported in both studies, the soundest interpretation from the K-Ar dating is perhaps that of McDougall. His statement that 'the age of the Pololu volcanic series is estimated not greater than 800,000 years, and probably considerably less than this' agrees with the paleomagnetic data, which also indicate an age less than 1 m.y.

Secular Variation in the Central Pacific Ocean

Current interest in geomagnetic secular variation of the central Pacific Ocean is due to two rather unusual features of the field in this area. The rate of change of the field has been anomalously low during the period covered by observatory measurements, as was noted by *Fisk* [1931]. Moreover, from a paleomagnetic study of Hawaiian lava flows which erupted between about 1750 and the present, we were recently able to demonstrate that this low rate of change in the direction of the field has persisted for at least some 200 years [*Doell and Cox,* 1963].

A second unusual characteristic of the earth's magnetic field in this area is that nondipole components of the field are greatly subdued at the present time. In a discussion of the low rate of secular variation and the supression of nondipole components, *Cox* [1962] suggested three possible explanations: (1) Sources in the core causing the nondipole irregularities, probably fluid eddies near the core-mantle boundary, never form under this region. (2) Sources of nondipole irregularities are present but are screened from observation at the surface by electrically conductive layers above the core-mantle boundary. (3) Sources are absent under

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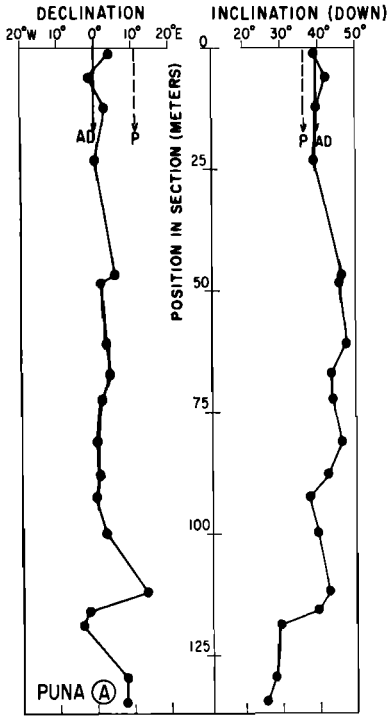


Fig. 15. Declination and inclination versus stratigraphic position for lavas of the Puna volcanic series. AD, axial dipole field values; P, present (inclined dipole) field values.

this region simply as a result of the random distribution of a small number of events (i.e. nondipole sources) over the surface of the core. If either (1) or (2) is the correct explanation, the anomalous characteristics of the field in this area will have persisted for a long time, so that presumably the paleomagnetic record of the geomagnetic field in Hawaii will be different from that of other areas. Conversely, if (3) is the correct explanation the Hawaiian paleomagnetic record should resemble those of comparable age from other regions.

The basic observational data used in studies of secular variation are the changes with time of the earth's magnetic field. For each of the sections studied paleomagnetically in Hawaii, we have presented a plot showing the average directions of magnetization of lava flows as a function of their vertical distance below some reference level (Figures 15 to 19). It is tempting to regard these plots as analogs of magnetograms with a time scale measured, perhaps, in centuries. Unfortunately, the resemblance to magnetograms is only superficial because the time interval between successive paleomagnetic datum points is unknown. Obtaining some idea of the time that elapsed between individual lava flows and between the formation of different stratigraphic sections is, in fact, the central problem in interpreting the paleomagnetic data from Hawaii in terms of secular variation of the earth's magnetic field.

Because we know so little about either the changes of the geomagnetic field over long periods of time or the rates of volcanism, there is considerable danger of circular reasoning when these two phenomena are compared for purposes of interpreting the paleomagnetic data in terms of secular variation. To avoid this, we will first attempt to set some limits to the time that elapsed during the accumulation of the different volcanic series on the basis of geologic evidence. We will then use this information to interpret the paleomagnetic data in terms of the history of the earth's magnetic field in the central Pacific region. We hope that this approach can be considerably strengthened in the near future by radiometric dating of the Hawaiian lavas.

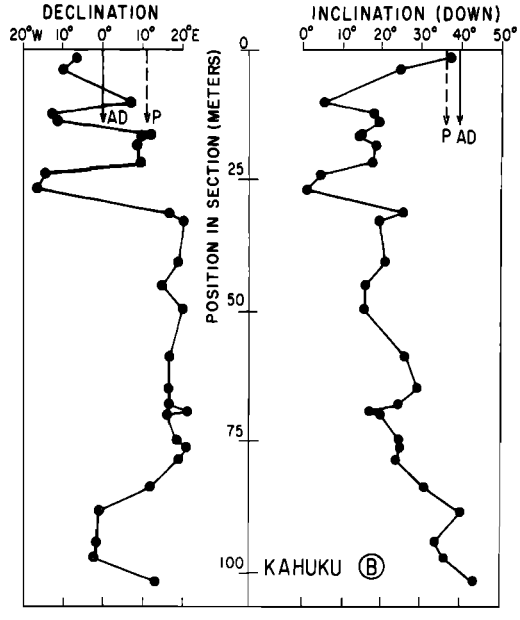


Fig. 16. Declination and inclination versus stratigraphic position for lavas of the Kahuku volcanic series. (See caption for Figure 15.)

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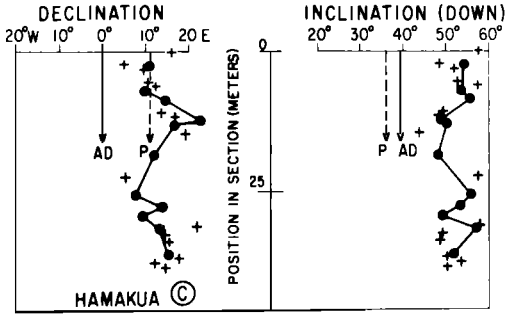


Fig. 17. Declination and inclination versus stratigraphic position for lavas of the Hamakua volcanic series. (See caption for Figure 15.)

Accumulation rates of volcanic series. Rates at which lava flows build up on Hawaii are highly variable. In calderas the rate may be extremely rapid; on the flanks of volcanoes geologic observations suggest a rather slow rate. A new flow on the side of a volcano usually rises above the surrounding slopes, and succeeding flows generally tend to flow in depressions between older flows. Thus two successive eruptions on the flanks of a volcano only rarely overlap. On the historically active volcanoes Mauna Loa and Kilauea, for example, the flows erupted during the last 200 years are sparsely distributed over a rather broad area and, except for those very near fissures or vents, one is rarely if ever superimposed on another.

The areal distribution of the post-Pahala ash flows, of the Puna and Kau volcanic series, on the still active volcanoes Kilauea and Mauna

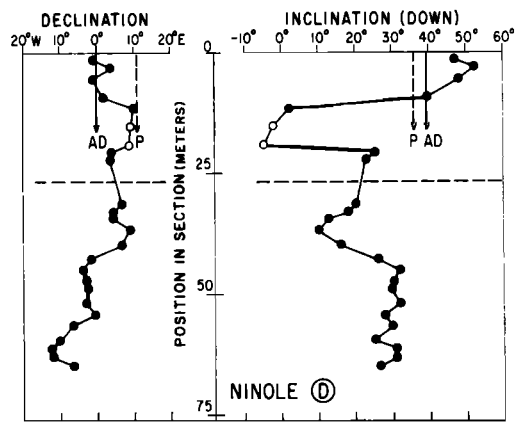


Fig. 18. Declination and inclination versus stratigraphic position for lavas of the Ninole volcanic series. (See caption for Figure 15.)

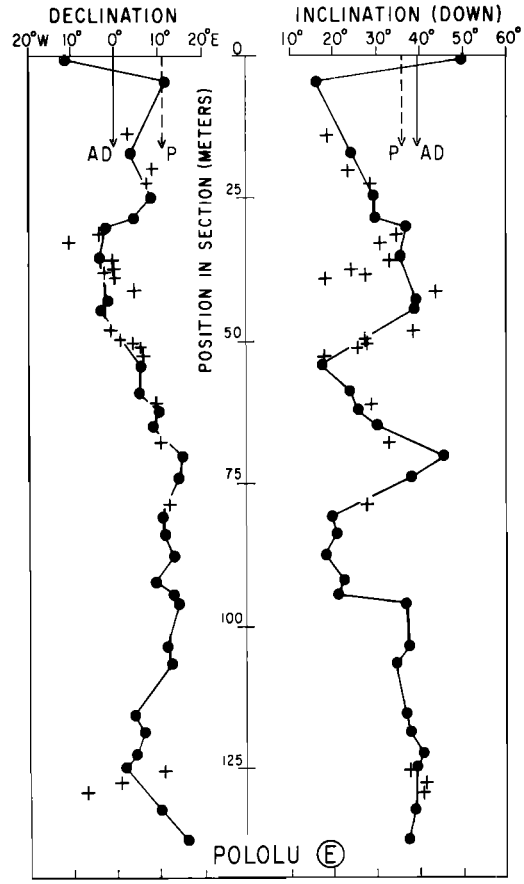


Fig. 19. Declination and inclination versus stratigraphic position for lavas of the Pololu volcanic series. (See caption for Figure 15.)

Loa may also be used for estimates of average accumulation rates. The Kau volcanic series is more than 180 meters thick in the walls of Mokuaweoweo, where 50 lavas were measured in 120 meters of section [*Stearns and MacDonald*, 1946, p. 190]. The Kau volcanic series thins to the southeast along the southeastern flank of Mauna Loa and terminates where the Pahala ash is exposed. Using *Rubin and Berthold*'s [1961] age of 10,000 years for the end of Pahala time, we calculate average extrusion rates varying from less than 130 years between successive flows in Mokuaweoweo to over 10,000 years on the southeast flank of Mauna Loa. Similarly, the Puna volcanic series ranges from a single flow on the southern flank of Kilauea to a sequence of 36 flows in 130 meters of measured section at

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Uwekahuna Bluff [*Stearns and MacDonald,* 1946, p. 193], where the base of the Puna volcanic series is not exposed. The average time between successive flows of the Puna volcanic series thus ranges from less than 270 years to about 10,000 years.

These relations suggest that for similar accumulations on the sides of volcanoes with similar rates of activity, the time between the emplacement of superimposed flows is rarely less than 200 years. Something like a thousand years may be a more realistic figure for the average time between superimposed flows.

Most of the flows studied paleomagnetically on Hawaii appear to have accumulated under conditions resembling those just described. One exception may be the Puna volcanic series, which in part contains massive columnar-jointed basalts several hundred feet thick, undoubtedly ponded in craters or in a rift-zone graben [*Stearns and MacDonald,* 1946]. However, the paleomagnetic samples were collected on Uwekahuna Bluff where the average flow thickness is 4 meters and the maximum is only 11, which suggests deposition on the carapace of the volcano some distance from the caldera. In the Puna volcanic series, where the time intervals between flows may have been less than elsewhere, alternate flows were sampled wherever possible. Every flow suitable for paleomagnetic purposes was sampled in the other series.

Under the assumption that an average of (*a*) 200 years and (*b*) 1000 years elapsed between superimposed flows, the following estimates may be made of the time represented in the different sections sampled:

Volcanic Series	(*a*) 200-Year Interval	(*b*) 1000-Year Interval
Puna	7,000 years	30,000 years
Kahuku	5,000	20,000
Hamakua	5,000	20,000
Ninole	5,000	20,000
Pololu	10,000	50,000

Undoubtedly the time represented in these sections is more variable than these figures suggest. However, recent K-Ar dating of volcanic series on other islands of the Hawaiian chain by *McDougall* [1964] suggests that these are reasonable order-of-magnitude estimates, and may even be low. For example, the Napali formation on Kauai accumulated over a time interval exceeding 1×10^6 years; the Waianae volcanic series on Oahu over a period of 7×10^5 years; and the Kula volcanic series on Maui over at least 4×10^5 years. Less complete sets of K-Ar dates from a number of other volcanic series on the Hawaiian chain suggest that they also accumulated over periods of several hundred thousand years. Thus, it is very unlikely that any of the Hawaiian volcanic series accumulated over periods of time less than thousands of years—they more probably represent time intervals of at least tens of thousands of years.

Puna volcanic series. The NRM directions of the two oldest flows, units 131 and 132, form a distinct group, as might be anticipated from the existence of an erosional unconformity above unit 132. After undergoing a moderate amount of variation, represented by units 133 to 135, the field remained remarkably close to the field of an axial dipole during the time required for the extrusion of the last 13 flows. As previously discussed, this series, being closer to the vent, may have accumulated at a faster rate than the other series. Nonetheless it is probable that the last 13 flows, which span more than 100 meters of section, correspond to an unusually long time interval during which the geomagnetic field remained essentially fixed in direction.

Kahuku volcanic series. Although not as thick as the Puna volcanic series, this series was extruded lower on the flanks of the volcano; moreover, thin ash layers occur between some flows, so that probably a longer time interval is represented than that for the Puna volcanic series. The earlier part of the paleomagnetic record in the Kahuku volcanic series is similar to that of the Puna volcanic series, although it is clearly older; the lowest flow has a direction about 10° from the axial dipole direction, after which the field moved near the dipole direction. The latter part of the Kahuku paleomagnetic record is strikingly different. The field remained close to a shallow northeasterly direction 30° from the direction of the axial dipole field during eruption of units 106 through 119 (57 meters aggregate thickness). The field then assumed a series of directions yet further removed from the dipole field, oscillating between shallow northeast and shallow northwest directions and finally returning to a nearly dipole direction when the topmost flow was extruded.

Hamakua volcanic series. All flows of the

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Hamakua volcanic series resemble the Puna volcanic series in having directions of magnetization in a tight group. This group is 15° steeper than the axial dipole field direction, which indicates that the geomagnetic field was steady but nonaxially dipolar during the entire time required for the extrusion of these 26 flows.

These paleomagnetic results clearly indicate that the Hamakua volcanic series cannot have been extruded during the same time interval as the Kahuku volcanic series. Thus they cannot be exact stratigraphic equivalents, although the geologic evidence indicates that they are not greatly different in age. During the time interval preceding deposition of the Pahala ash, when these two series were extruded, it appears that the geomagnetic field remained for long intervals of time in several nonaxially dipolar positions.

Ninole volcanic series. During the time required for the formation of the Ninole volcanic series, the geomagnetic field generally behaved as it did during extrusion of the Kahuku volcanic series. The oldest 11 flows (units 51 through 61, representing an aggregate thickness of 25 meters) have NRM directions tightly grouped away from the axial dipole direction. The field then moved somewhat irregularly toward shallow northeast directions (there is a stratigraphic break of unknown duration separating units 66 and 67), even becoming negatively inclined at one time, and finally returned to a direction slightly steeper than that of an axial dipole. During the time required for the extrusion of only 5 flows (units 70 through 74), the geomagnetic field changed direction by 55°.

Pololu volcanic series. This series is 138 meters thick and occurs on the flanks of Kohala volcano, so that it probably represents a longer time interval than any of the other sections. It is not surprising, therefore, that many of the features seen in the other sections are also present here—long periods with relatively little variation (units 1 through 12), as well as periods of more marked changes (units 20 through 32). However, there is also a rather interesting overall feature in that the field appears to have oscillated between two groups of directions. The first has an inclination of about 40°, nearly that of an axial dipole, with declinations ranging between 355 and 15°. The second group has inclinations in the range 30 to 15° and declinations between 0 and 15°. During extrusion of the lowest 12 flows, the geomagnetic field was in the more nearly dipolar direction. Subsequently it oscillated several times between this group and the more shallowy inclined one. Quite possibly this represents an oscillation of the geomagnetic field between two stable directions, but gaps in the stratigraphic record would cause similar variations.

Discussion. Time intervals between eruption of successive flows in the Hawaiian volcanic series undoubtedly varied considerably. However, it is very unlikely that all groups of successive flows that indicate marked changes of the geomagnetic field were related to long intraflow time intervals and that groups indicating constant field directions were extruded during short time intervals. As was mentioned earlier, these time intervals are probably of the order of at least several tens of thousands of years. Steady directions of the geomagnetic field over such long intervals of time implies a marked reduction in the nondipole field, which fluctuates with periods of 1000 years and less [*Cox and Doell,* 1964]. During some time intervals the main dipole field remained constant, sometimes aligned parallel to the axis of rotation, sometimes at marked angles to it. During other time intervals of similar duration, much larger changes in the direction of the field occurred. These changes may have been caused either by changes in the orientation of the main dipole or by the sporadic reappearance of the nondipole field.

We turn now to the problem of a physical explanation of the attenuation of a nondipole field in the central Pacific. The general, and perhaps complete, absence of this feature during the last several hundreds of thousands of years clearly eliminates the possibility that it is due to an unusual asymmetric distribution of the small number of rapidly changing sources that produce the field. The present paleomagnetic data offer no basis for choosing between the two other possible explanations [*Cox,* 1962].

However, we have recently considered this problem in more detail in a study of the longer-period variations of the geomagnetic field [*Cox and Doell,* 1964]. Briefly, we concluded that shielding by electrically conducting layers in the mantle beneath the central Pacific is physically quite difficult and that the best explanation for the absence of the nondipole field in this

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region is the absence of sources. We have also suggested a model, involving temperature differences in the lower mantle, that could account for the general absence of fluid eddies which are probably the cause of the nondipole field at the surface of the core under the Pacific.

The absence of a nondipole field in the central Pacific permits a much more detailed paleomagnetic analysis of main dipole changes than may be made in other regions. The major groups of directions in the Hawaii paleomagnetic data probably correspond to major shifts of the dipole axis, some shifts being as large as 15°. If our conclusions about volcanic extrusion rates are correct, this dipole wobble involves periods of at least several tens of thousands of years.

Finally, we should like to consider the fact that the average orientation of the dipole field throughout the extrusion of the Hawaiian lavas is displaced some 5° from the rotation axis. As discussed earlier, the confidence circle drawn about the average direction is meaningless (Figure 13*b*) except as a lower limit to the true confidence limit. However, additional analyses were made after rejecting all flows with 95% confidence angles >8°, >7°, and >6° and after including only data from every third flow. The mean positions of the poles as calculated from these analyses (including that for all flows) were all within 1° of each other. These analyses indicate a statistical stability for the Hawaiian data, even if confidence limits cannot be rigorously determined. If we include data from lavas we have collected in the western United States, Alaska, and the Pribilof Islands (all <1 m.y. old) in an analysis of the Hawaiian data, the number of data points is doubled, the average VGP position is the same as for the Hawaiian data alone, and the statistical parameters are essentially unaltered [*Cox and Doell,* 1964]. It seems, therefore, that the 5° displacement between the average dipole axis during the last 1 million years and the present rotation axis is real. Whether this displacement is due to a shift in the axis of rotation or to an average displacement between the rotation and geomagnetic dipole axis over the last million years cannot be determined from the present paleomagnetic data.

Acknowledgments. We wish to thank Messrs. T. Robinson, E. Roth, and G. Anderson for assistance with the magnetic measurements; Mr. W. LeMasurier for the petrographic examinations; and Mr. G. Dakujaku for assistance in collecting samples.

Partial support under National Science Foundation grant GF-77 is gratefully acknowledged.

Publication authorized by the Director, U. S. Geological Survey.

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(Manuscript received March 15, 1965.)

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